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Exposé V

ORDINARY CRYSTALS AND CANONICAL COORDINATES

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CONTENTS

0. INTRODUCTION

Throughout this exposé, k denotes an algebraically closed field of characteristic $p > 0$, and W the ring of Witt vectors over k.

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0. INTRODUCTION

Let X_0/k be an ordinary elliptic curve. By Serre-Tate theory [14, V], the formal moduli space M of deformations of X_0 over local Artinian W-algebras with residue field k is isomorphic to the formal multiplicative group $\widehat{G}_m/W$. More precisely, the p-divisible group

$$G_0 = \cup\, \mathrm{Ker}(p^n : X_0 \to X_0)$$

is isomorphic to $(\mathbb{Q}_p/\mathbb{Z}_p) \times \mu_{p^\infty}$, and the choice of an isomorphism α between the étale part of G_0 and $\mathbb{Q}_p/\mathbb{Z}_p$ canonically identifies, for every local Artinian W-algebra R with residue field k, the set of liftings of X_0 over R with the group $\mathrm{Ext}^1((\mathbb{Q}_p/\mathbb{Z}_p)_R, (\mu_{p^\infty})_R)$, which in turn is canonically isomorphic to the group of units of R congruent to 1 modulo the maximal ideal. Thus one obtains an isomorphism, depending only on α, between M and $(\widehat{G}_m)_W$; in particular, the universal deformation X of X_0 over M defines a "canonical coordinate" q on M such that $M \simeq \mathrm{Spf}\, W[[q-1]]$. The parameter q, which describes the p-divisible group G associated with X as an extension of $\mathbb{Q}_p/\mathbb{Z}_p$ by μ_{p^∞}, also admits another interpretation in terms of the module $H^1_{\mathrm{DR}}(X/M)$, equipped with its F-crystal structure and its Hodge filtration $H^0(X, \Omega^1_{X/M}) \subset H^1_{\mathrm{DR}}(X/M)$. The datum of an isomorphism α as above gives, by duality, an isomorphism α' between the formal group $\widehat{X}/M$ associated with X and the formal group $(\widehat{G}_m)_M$; hence it gives a differential form $b \in H^0(X, \Omega^1_{X/M})$ corresponding under α' to the invariant form on $(\widehat{G}_m)_M$. Let a $\in H^1_{\mathrm{DR}}(X/M)$ satisfy $\nabla a = 0$, and let (a, b) be a symplectic basis of $H^1_{\mathrm{DR}}(X/M)$, i.e. $\langle a, b\rangle = 1$ and $\langle a, a\rangle = 0$. If

$$\nabla : H^1_{\mathrm{DR}}(X/M) \to \Omega^1_{M/W} \otimes H^1_{\mathrm{DR}}(X/M)$$

denotes the Gauss-Manin connection, then necessarily

$$(0.1) \qquad \nabla b = \eta \otimes a,$$

with $\eta \in \Omega^1_{M/W}$. One can show (see the Appendix and the following Exposé, by N. Katz) that

$$(0.2) \qquad \eta = d \log q,$$

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where q is the parameter defined above, and hence coincides with the parameter independently defined by Dwork ([8], [6]) using (0.2). The basis (a, b) above also has remarkable properties with respect to the Frobenius operation F on $H^1_{\mathrm{DR}}(X/M)$: if one chooses, as the endomorphism of M lifting Frobenius on $M \otimes k$, the endomorphism given by $q \mapsto q^p$, then F is expressed by

$$(0.3) \qquad F\, a = a, \qquad F\, b = p\, b.$$

The preceding constructions generalize to ordinary abelian varieties [8]; see also [9], [11] for arithmetic applications in the case of elliptic curves.

The purpose of this exposé is to show that an analogous theory can be developed for ordinary K3 surfaces, at least when $p \neq 2$. Let X_0/k be an ordinary K3 surface, i.e. one for which Frobenius on $H^2(X_0, \mathcal{O})$ is nonzero, and let S be the formal moduli space of deformations of X_0 over local Artinian W-algebras with residue field k (IV 1.2). If $p \neq 2$, we prove that S is canonically endowed with a formal group structure isomorphic to $(\widehat{G}_m)^{20}_W$. In particular, the universal deformation X of X_0 over S defines "canonical coordinates" q_i $(1 \leq i \leq 20)$ such that

$$S \simeq \operatorname{Spf} W[[q_1 - 1, \ldots, q_{20} - 1]],$$

and these form a basis of the characters of S. We further show that, if L_0 is a nontrivial invertible sheaf on X_0, the hypersurface $\Sigma(L_0)$ of S over which L_0 extends to $X \times_S \Sigma(L_0)$ (IV 1.5) is the kernel of a character of S, defined in an almost tautological way by the crystalline Chern class of L_0.

In the absence of a Serre-Tate theory for liftings of K3 surfaces, the formal group structure on S is defined in terms of the F-crystal structure on $H^2_{\mathrm{DR}}(X/S)$. In fact, the case of abelian varieties and that of K3 surfaces can be included in a single structure theorem for a certain class of ordinary F-crystals. The study of these crystals is the subject of

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No. 1. The geometric applications are given in No. 2. As regards the theorem concerning the hypersurfaces $\Sigma(L_0)$, the essential ingredient is the use of the crystalline Chern class to control the obstruction to extension.

1. CANONICAL PARAMETERS OF ORDINARY HODGE F-CRYSTALS

1.1. Review of definitions.

1.1.1. For the definitions and general properties of F-crystals, see Katz [8], [10]. Throughout No. 1 we fix the formal power-series rings

$$A_0 = k[[t_1, \ldots, t_n]], \qquad A = W[[t_1, \ldots, t_n]],$$

where $(t_1, \ldots, t_n)$ is a finite sequence of indeterminates (for $n = 0$, we take $A_0 = k$ and $A = W$). A crystal over A_0 (understood to be in finite free modules) is, by definition, a finite free A-module H equipped with a connection

$$\nabla : H \to \Omega^1_{A/W} \otimes H$$

that is integrable and p-adically topologically nilpotent. Integrability of ∇ means that, on putting $D_i = d/\mathrm{dt}_i$, one has

$$\nabla(D_i)\nabla(D_j) = \nabla(D_j)\nabla(D_i)$$

for all $i \neq j$. This makes the PD-differential operators on A, i.e. the polynomials in the D_i with coefficients in A, act on H by the formula

$$\nabla(\textstyle\sum_m a_m D^m) = \sum_m a_m (\nabla(D))^m,$$

with the usual condensed notation $D^m = D_1^{m_1} \cdots D_n^{m_n}$, etc. The topological nilpotence of ∇ means that, for every i, $\nabla(D_i)^m$ tends to zero in the p-adic topology of $\mathrm{End}_W(H)$ as m tends to $+\infty$. By Berthelot [1, II 4], the

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datum of a connection ∇ as above is equivalent to the following datum: for every pair of W-homomorphisms (f, g) from A to a local, Noetherian, complete W-algebra B, such that f and g are congruent modulo an ideal I of B endowed with divided powers compatible with the standard divided powers of p, an isomorphism

$$\chi(f, g) : f^*H \simeq g^*H, \qquad (*)$$

subject to the transitivity conditions made explicit in [8, 1.2], namely $\chi(g, h)\chi(f, g) = \chi(f, h)$, $\chi(\mathrm{fk}, \mathrm{gk}) = k^*\chi(f, g)$, and $\chi(\mathrm{id}, \mathrm{id}) = \mathrm{id}$. The following formula describes concretely the part of this dictionary that we need.

LEMMA 1.1.2. With the notation above, let

$$f^* : H \to f^*H$$

denote the adjunction map, where $f^*H = f_*f^*H$ by abuse of notation. The composite A-linear homomorphism

$$\chi(f, g)f^* : H \xrightarrow{f^*} f^*H \xrightarrow{\chi(f, g)} g^*H$$

is given by

$$(1.1.2.1) \qquad \chi(f, g)f^*(x) = \sum_m (f(t) - g(t))^{[m]}\, g^*(\nabla(D)^m x),$$

where the sum ranges over all multi-indices $m = (m_1, \dots, m_n) \in N^n$, with the condensed notation

$$(f(t) - g(t))^{[m]} = (f(t_1) - g(t_1))^{[m_1]} \cdots (f(t_n) - g(t_n))^{[m_n]}$$

(the brackets denote a divided power in I), and

$$\nabla(D)^m = \nabla(D_1)^{m_1} \cdots \nabla(D_n)^{m_n}.$$

(N.B. The series on the right-hand side of (1.1.2.1) converges by the topological nilpotence of ∇.)

This statement is essentially contained in [1, II 4.3]. It suffices to verify it after reduction modulo p^r for every r. We shall again write

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(*) We also write f (respectively g) : Spec$(B) \to$ Spec(A) for the corresponding morphism of schemes.

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by abuse of notation $(f, g) : A \rightrightarrows B$ for the W_r-homomorphisms obtained from the given homomorphisms by reduction modulo p^r. Let D be the divided-power envelope of the augmentation ideal J of $A \widehat{\otimes}_{W_r} A$ $(= W_r[[t_1, \dots, t_n, u_1, \dots, u_n]])$, and write d_0 (respectively d_1) $: A \to D$ for the homomorphism given by a $\mapsto$ a $\otimes$ 1 (respectively 1 $\otimes$ a), defining the left (respectively right) A-module structure on D. Since J, generated by the $t_i - u_i$, is regular, it follows from [1, I 4.5.1] that, for the left structure, D identifies with the divided-power algebra

$A\langle \xi_1, \dots, \xi_n \rangle$,

where $\xi_i = d_1 t_i - d_0 t_i$ is the image of $u_i - t_i$ in D. By the universal property of divided-power envelopes, and because B is complete, there is a PD-morphism $s : D \to B$ such that $s\,d_0 = g$ and $s\,d_1 = f$. On the other hand, by [1, II 4.3.8], the connection ∇ defines a PD-stratification

$$\varepsilon : D \otimes_A H \,(= d_1^* M) \simeq H \otimes_A D \,(= d_0^* M)$$

such that the d_1-linear composite

$$\theta : H \xrightarrow{d_1^*} d_1^* H \xrightarrow{\varepsilon} d_0^* H$$

is given by

$$(*) \qquad \theta(x) = \sum_m \nabla(D^m)x \otimes \xi^{[m]},$$

with the evident condensed notation. By definition, $\chi(f, g)$ is the isomorphism obtained from ε by pullback along s:

$$\chi(f, g) = s^* \varepsilon : f^* H \simeq g^* H.$$

If f (respectively g) is regarded as defining a right (respectively left) A-module structure on B, there is a commutative diagram

$$\begin{array}{ccccc} H & \xrightarrow{d_1^*} & D \otimes_A H & \xrightarrow{\varepsilon} & H \otimes_A D \\ \| & & \downarrow s \otimes 1 & & \downarrow 1 \otimes s \\ H & \xrightarrow{f^*} & B \otimes_A H & \xrightarrow{\chi(f,g)} & H \otimes_A B, \end{array}$$

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which shows, in view of (*), that the lower horizontal composite is given by

$$\chi(f,g)f^*(x) = \sum_m \nabla(D)^m x \otimes s(\xi)^{[m]}.$$

But

$$s(\xi_i) = s(d_1 t_i - d_0 t_i) = f(t_i) - g(t_i),$$

and therefore

$$\begin{aligned}\chi(f,g)f^*(x) &= \sum_m \nabla(D)^m x \otimes (f(t)-g(t))^{[m]} \\ &= \sum_m (f(t)-g(t))^{[m]}\, g^*(\nabla(D)^m x).\end{aligned}$$

1.1.3. An F-crystal over A_0 is, by definition, a crystal (H,∇) over A_0 equipped, for every lifting $\varphi : A \to A$ of Frobenius on A_0 compatible with the Frobenius endomorphism σ of W, with a horizontal homomorphism

$$F(\varphi) : \varphi^* H \to H$$

($\varphi^* H$ being equipped with the connection $\varphi^*\nabla$), such that $F(\varphi) \otimes \mathbb{Q}_p$ is an isomorphism and such that, for every pair of liftings (φ,ψ), there is a commutative triangle

(1.1.3.1)

$$\begin{tikzcd} \varphi^* H \arrow[r, "F(\varphi)"] \arrow[d, "\chi(\varphi,\psi)"'] & H \\ \psi^* H, \arrow[ur, "F(\psi)"'] & \end{tikzcd}$$

where $\chi(\varphi,\psi)$ is the canonical isomorphism defined by ∇, using the fact that φ is congruent to ψ modulo the divided-power ideal pA.

The horizontality of $F(\varphi)$ is expressed by the commutativity of the square

(1.1.3.2)

$$\begin{tikzcd} \varphi^* H \arrow[r, "\varphi^*\nabla"] \arrow[d, "F(\varphi)"'] & \Omega^1_{A/W} \otimes \varphi^* H \arrow[d, "1 \otimes F(\varphi)"] \\ H \arrow[r, "\nabla"] & \Omega^1_{A/W} \otimes H. \end{tikzcd}$$

Since, by definition, $\varphi^*\nabla$ makes the square

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$$\begin{array}{ccc} H & \xrightarrow{\nabla} & \Omega^1_{A/W} \otimes H \\ \downarrow \varphi^* & & \downarrow \varphi^* \otimes \varphi^* \\ \varphi^* H & \xrightarrow{\varphi^* \nabla} & \Omega^1_{A/W} \otimes \varphi^* H \end{array}$$

commute, one obtains the formula

$$(1.1.3.3) \qquad \nabla F(\varphi)\varphi^* x = \sum_i \varphi^* \omega_i \otimes F(\varphi)\varphi^* h_i,$$

for $x \in H$ such that $\nabla x = \sum_i \omega_i \otimes h_i$.
On the other hand, the variation law for $F(\varphi)$ in (1.1.3.1) is expressed by the formula

$$(1.1.3.4) \qquad F(\psi)\psi^* x = F(\varphi)\varphi^* x + \sum_{|n|>0} (\psi(t) - \varphi(t))^{[n]} F(\varphi)\varphi^*(\nabla(D)^n x),$$

with the notation of 1.1.2. Indeed, from the transitivity conditions satisfied by χ,

$$\begin{aligned} F(\psi)\psi^* x &= F(\psi)\chi(\varphi,\psi)\chi(\psi,\varphi)\psi^* x \\ &= F(\varphi)\chi(\psi,\varphi)\psi^* x \qquad (1.1.3.1), \end{aligned}$$

whence 1.1.3.4, by 1.1.2.1 applied with $f = \psi$ and $g = \varphi$.
1.1.4. Let $s_0 : A_0 \to k'$ be a point of Spec(A_0) with values in an algebraically closed extension of k. If φ lifts Frobenius as above, it is known [8, 1.1] that there is a unique lifting of s_0 to $s : A \to W(k')$ such that $s\varphi = \sigma s$ (where σ denotes Frobenius on $W(k')$); this is called the Teichmüller lifting of s_0 relative to φ. Let H be an F-crystal over A_0. Then $F(\varphi)$ gives a $\sigma-$linear endomorphism of $s^* H$, and hence an F-crystal $s^* H$ over k'. If φ' is another lifting of Frobenius, and s' is the corresponding Teichmüller lifting of s_0, the F-crystals $s^* H$ and $s'^* H$ are canonically isomorphic by (1.1.3.1) (cf. [8, 1.4]). We shall write $s_0^* H$ for $s^* H$, and say that $s_0^* H$ is the F-crystal induced by H at the point s_0.
1.1.5. Finally, recall that to every F-crystal over k one associates a Newton polygon and a Hodge polygon; for their definitions and properties we refer the reader to [12], [8], and [10].

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1.2. Unit F-crystals.

1.2.1. An F-crystal (H, ∇) over A_0 is called a unit F-crystal if, for a lifting φ to A of Frobenius on A_0, $F(\varphi)$ is an isomorphism. By (1.1.3.1), this is equivalent to requiring $F(\varphi)$ to be an isomorphism for every lifting φ.

Every finite free $\mathbb{Z}_p-$module M gives a unit F-crystal over A_0, namely $(H = M \otimes A,\ \nabla = 1 \otimes d,\ F(\varphi) = 1 \otimes \varphi)$. In fact, every unit F-crystal over A_0 is of this form:

PROPOSITION 1.2.2 (cf. [8, §3]). Let H be a unit F-crystal over A_0. There exists a basis (e_i) $(1 \leq i \leq r)$ of H over A such that $\nabla e_i = 0$ and $F(\varphi)\varphi^* e_i = e_i$ for every lifting φ of Frobenius on A_0.

In other words, the basis (e_i) defines an isomorphism from the trivial F-crystal $\mathbb{Z}_p^r \otimes A$ defined above onto H.

Before proving 1.2.2, let us first make two remarks.

$a)$ If $x \in H$ is such that, for some lifting φ, $F(\varphi)\varphi^* x = x$, then $\nabla x = 0$. Indeed, since φ lifts Frobenius,

$$\varphi^*(\Omega^1_{A/W}) \subset p\Omega^1_{A/W};$$

hence p divides $\nabla x = \nabla F(\varphi)\varphi^* x$ by (1.1.3.3). The same argument shows inductively that ∇x is divisible by p^n for every n, and therefore $\nabla x = 0$.

$b)$ If $x \in H$ satisfies $F(\varphi)\varphi^* x = x$ for one lifting φ, then $F(\psi)\psi^* x = x$ for every lifting ψ. In view of a), this follows from the variation law for $F(\varphi)$, (1.1.3.4).

It is therefore enough to find a basis (e_i) of H over A such that $F(\varphi)\varphi^* e_i = e_i$ for one chosen lifting φ, for example the one given by $t_j \mapsto t_j^p$ $(1 \leq j \leq n)$. Let $s : A \to W$ be the augmentation given by $s(t_i) = 0$. Since $s\varphi = \sigma s$, $F(\varphi)\varphi^*$ induces on $s^* H$ a $\sigma-$linear automorphism Φ. Since k is algebraically closed, there exists a basis of $s^* H$ fixed by Φ (start with a basis fixed by Φ modulo p, which exists by Lang's lemma, and modify it successively so as

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to make it fixed by Φ modulo p^n for every n). In other words, there exists a basis (f_i) $(1 \le i \le r)$ of H such that, on writing $F(\varphi)\varphi^* = F$, one has

Ff = (I+M)f,

with $M \in M_r(A)$ and $M = 0$ mod(t) (= $(t_1, \dots, t_n)A$). It is enough to show that f may be replaced by e = (I+N)f, where $N \in M_r(A)$ and $N = 0$ mod(t), so that Fe = e. Written out, this gives the equation

$$(*) \qquad N = M + \varphi(N) + \varphi(N)M.$$

The map $N \mapsto M + \varphi(N) + \varphi(N)M$ is contractive for the (t)-adic topology, since $\varphi((t)) \subset (t)^p$. Thus (*) has a unique solution, completing the proof of 1.2.2.

1.2.3. We shall say that an F-crystal (H, ∇) over A_0 is topologically nilpotent if, for a lifting φ of Frobenius on A_0, $F(\varphi)\varphi^*$ is p-adically topologically nilpotent. This condition is independent of the choice of φ, as follows from (1.1.3.4) (or, more precisely, from the analogue of (1.1.3.4) for iterates of $F(\varphi)\varphi^*$ and $F(\psi)\psi^*$). It is not true that every F-crystal over A_0 is an extension of a topologically nilpotent F-crystal by a unit F-crystal. More precisely, one has the following criterion, a special case of a theorem of Katz [10, 2.4.2]:

PROPOSITION 1.2.4. Let H be an F-crystal over A_0. Let F_0 denote the endomorphism of $H_0 = H \otimes_A A_0$ induced by $F(\varphi)\varphi^*$, where φ lifts Frobenius F_{A_0} on A_0 (F_0 is F_{A_0} − linear and independent of φ). The following conditions are equivalent:

(i) The stable ranks of (H_0, F_0) at the closed point and at an algebraically closed geometric point lying over the generic point of Spec(A_0) are equal.

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(ii) There exists an extension of F-crystals over A_0,

$$0 \to U \to H \to E \to 0,$$

where U is a unit F-crystal and E is a topologically nilpotent F-crystal.

Recall what is meant by “stable rank.” If L is a finite-dimensional vector space over a perfect field of characteristic p, equipped with a p-linear endomorphism Φ, then L has a canonical decomposition

$$L = L^{\mathrm{ss}} \oplus L^{\mathrm{nilp}},$$

where $L^{\mathrm{ss}} = \cap \operatorname{Im} \Phi^n$ and $L^{\mathrm{nilp}} = \cup \operatorname{Ker} \Phi^n$; the dimension of L^{ss} is, by definition, the stable rank of (L, Φ).

Although this is a special case of [10, 2.4.2], let us briefly give a proof of 1.2.4 (*), since the present situation is distinctly simpler than that of (loc. cit.). It is clear that (ii) implies (i). Conversely, put $\overline{H}_0 = H_0 \otimes_{A_0} k = H \otimes_A k$. Lift to H_0 a basis of $\overline{H}_0$ adapted to the decomposition of $\overline{H}_0$ under $F_0 \otimes k$,

$$\overline{H}_0 = \overline{H}_0^{\mathrm{ss}} \oplus \overline{H}_0^{\mathrm{nilp}}.$$

Then F_0 is represented by a matrix
(a c)
$(b\ d)$,
with $b = 0$ mod(t), and a invertible of rank $r = \dim \overline{H}_0^{\mathrm{ss}}$. In a new basis of H_0 given by the change-of-basis matrix

$$P = (1_r \quad 0)$$
$$(x \quad 1),$$

the matrix of F_0 is

$$(a' \quad c') \qquad (a \quad c)$$
$$(b' \quad d') = P^{-1} \quad (b \quad d)\, P^{(p)},$$

and $x = 0$ mod(t) can be chosen so that $b' = 0$. Indeed, x must satisfy $x =$ f(x), where

$$f(x) = b\,a^{-1} - \mathrm{xcx}^{(p)} a^{-1} + d\,x^{(p)} a^{-1}.$$

This equation has a unique solution because f is contractive for the (t)-adic topology. In this new basis, the matrix of F_0 therefore has the form

$$(a' \quad c')$$
$$(0 \quad d'),$$

with a' invertible of rank r. Hypothesis (i) then implies that the endomorphism d' is nilpotent, because this is true after localizing d' in an algebraically closed extension of the fraction field of A_0. Choose a lifting φ of F_{A_0}. In a basis of H lifting the basis of H_0

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(*) appearing in unpublished notes of Katz predating (loc. cit.).

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chosen above, the matrix of $F = F(\varphi)\varphi^*$ is written (with changed notation)
(a c)
(b d),
where $b = 0$ mod p and d is topologically nilpotent. Applying the fixed-point method once again (now with a map contractive for the p-adic topology), one sees that a new basis $(e_1, \dots, e_r, e_{r+1}, \dots, e_{r+s})$ of H can be found, given by a change-of-basis matrix

$$\begin{pmatrix} 1_r & 0 \\ x & 1_s \end{pmatrix},$$

with $x = 0$ mod p, such that in this new basis the matrix of F is

$$\begin{pmatrix} a' & c' \\ 0 & d' \end{pmatrix},$$

where a' is invertible (of rank r) and d' is topologically nilpotent. The sub-F-module

$$U = \oplus_{1 \le i \le r} \mathrm{Ae}_i$$

is equal to $\cap_{n \ge 0}$ Im F^n and hence, by (1.1.3.2), is horizontal, i.e.

$$\nabla U \subset \Omega^1_{A/W} \otimes U.$$

By (1.1.3.1), U is therefore a unit sub-F-crystal of H, and by construction the F-crystal H/U is topologically nilpotent. This completes the proof.

REMARK 1.2.5. Under the hypotheses of 1.2.4, the F-crystal U is characterized, in terms of H, by $U = \cap$ Im F^n (where $F = F(\varphi)\varphi^*$). We shall call U the unit sub-F-crystal of H.

1.3. Ordinary F-crystals.

1.3.1. Let H be an F-crystal over A_0, and set $H_0 = H \otimes_A A_0$. Let φ be a lifting to A of Frobenius F_{A_0}, and put

$$H_0^{(p)} = F^*_{A_0} H_0 = \varphi^* H \otimes_A A_0.$$

For $i \in \mathbb{N}$, define

$$(1.3.1.1) \qquad M^i H_0^{(p)} = \{x \in H_0^{(p)} \,|\, \text{there exists } y \in \varphi^* H \text{ lifting } x \text{ such that } F(\varphi)\, y \in p^i H\},$$

$$(1.3.1.2) \qquad \mathrm{Fil}^i H_0 = H_0 \cap M^i H_0^{(p)}$$

(where H_0 is regarded as a submodule of $F_{A_0 *} H_0^{(p)}$ through the adjunction map),

$$(1.3.1.3) \qquad \mathrm{Fil}_i H_0 = \{x \in H_0 \,|\, \text{there exists } y \in H \text{ lifting } x \text{ such that } p^i y \in \text{ Im } F(\varphi)\}.$$

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By (1.1.3.1), the submodules $\mathrm{Fil}^i H_0$ (respectively $\mathrm{Fil}_i H_0$) do not depend on the choice of φ; they form a decreasing (respectively increasing) filtration of H_0, called the Hodge filtration (respectively the conjugate filtration), cf. [15, 1.9], where this filtration is denoted F^i_{Hodge} (respectively F^{-i}_{con}). One can also show [15, 2.2] that

$$(1.3.1.4) \qquad \mathrm{Fil}^i H_0 = \{x \in H_0 \,|\, \text{there exists } y \in H \text{ lifting } x \text{ such that } F(\varphi)\varphi^* y \in p^i H\}.$$

The terminology comes from the Mazur-Ogus theorem [3, 8.26], which asserts that if X_0/A_0 is a proper smooth scheme (or formal scheme) such that

(i) the crystalline cohomology $H^*(X_0/A)$ is a free A-module,

(ii) the spectral sequence

$$E_1^{\mathrm{ij}} = H^j(X_0, \Omega^i_{X_0/A_0}) \Rightarrow H^*_{\mathrm{DR}}(X_0/A_0)$$

degenerates at E_1 and E_1 is free over A_0 and compatible with base change, then, on putting $H = H^n(X_0/A)$ (and hence $H_0 = H^n_{\mathrm{DR}}(X_0/A_0)$), with n fixed, the filtration $\mathrm{Fil}^i H_0$ (respectively $\mathrm{Fil}_i H_0$) coincides with the canonical filtration on the abutment of the preceding spectral sequence (respectively of the conjugate spectral sequence

$$E_2^{\mathrm{ij}} = H^i(X_0, H^j(\Omega^\bullet{}_{X_0/A_0})) \Rightarrow H^*_{\mathrm{DR}}(X_0/A_0)).$$

Recall [15, 1.6] that the filtrations 1.3.1.2 and 1.3.1.3 are finite, separated, and exhaustive, and that for a given n the conditions $\mathrm{Fil}^{n+1} H_0 = 0$ and $\mathrm{Fil}_n H_0 = H_0$ are equivalent; one then says that H has level $\leq n$.

We shall write

$$(1.3.1.5) \qquad \mathrm{gr}^i H_0 \qquad (\text{respectively } \mathrm{gr}_i H_0)$$

for the graded module associated with the filtration $\mathrm{Fil}^i H_0$ (respectively $\mathrm{Fil}_i H_0$). If $\mathrm{gr}_i H_0$ is free over A_0, then so is $\mathrm{gr}^i H_0$; formation of $\mathrm{gr}_i H_0$ and $\mathrm{gr}^i H$ commutes with base change; and for every i, $\mathrm{gr}_i H_0$ and $\mathrm{gr}^i H_0$ are canonically isomorphic. In particular they have the same rank h_i, called the i-th Hodge number of H [15, 1.12, 2.3].

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Recall also [15, 1.6, 2.6] that the conjugate filtration is horizontal for the connection ∇_0 induced on H_0 (and has associated graded object of zero p-curvature), whereas the Hodge filtration satisfies Griffiths transversality:

$$(1.3.1.6) \qquad \nabla_0 \operatorname{Fil}^i H_0 \subset \Omega^1_{A_0/k} \otimes \operatorname{Fil}^{i-1} H_0.$$

The notion of an ordinary F-crystal was isolated by Mazur [12] and Ogus, to whom the following characterization is due [15, 3.1.3]:

PROPOSITION 1.3.2. Let H be an F-crystal over A_0 such that $\mathrm{gr}_\bullet H_0$ is free over A_0. The following conditions are equivalent:

$$(i)\ \text{the Newton and Hodge polygons of the } F-\text{crystal } e_0^* H \text{ induced by } H \text{ at the closed point } e_0 : A_0 \to k\ (1.1.4) \text{ coincide;}$$

(i') for every point s_0 of $\mathrm{Spec}(A_0)$ with values in an algebraically closed extension k' of k, the Newton and Hodge polygons of the F-crystal $s_0^*(H)$ (1.1.4) coincide;

(ii) the Hodge and conjugate filtrations of H_0 are opposite, i.e. for every i,

$$H_0 = \operatorname{Fil}_i H_0 \oplus \operatorname{Fil}^{i+1} H_0;$$

(iii) there exists a unique filtration of H by sub-F-crystals

$$(1.3.2.1) \qquad 0 \subset U_0 \subset U_1 \subset \ldots \subset U_i \subset U_{i+1} \subset \ldots$$

such that U_i lifts $\operatorname{Fil}_i H_0$ and U_i/U_{i-1} is of the form $V_i(-i)$, where V_i is a unit F-crystal and $(-i)$ denotes the Tate twist that replaces F by $p^i F$.

DEFINITION 1.3.3. An F-crystal H over A_0 is called ordinary if $\mathrm{gr}_\bullet H_0$ is free and H satisfies the equivalent conditions of 1.3.2.

Let us prove 1.3.2. The implication $(i') \Rightarrow (i)$ is immediate, and it is clear that (iii) implies (i'). We prove $(i) \Rightarrow$ (ii). Suppose H has level $\leq n$ and that the conclusion is known for F-crystals of level $\leq n-1$. Let F denote the p-linear endomorphism of H_0 induced

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by $F(\varphi)$ for a lifting φ (F does not depend on this choice). By definition (1.3.1),

$$(1.3.3.1) \qquad \mathrm{Fil}_0\, H_0 = \mathrm{Im}\, F : H_0^{(p)} \to H_0, \qquad \mathrm{Fil}^1\, H_0 = \mathrm{Ker}\, F : H_0 \to H_0.$$

Hypothesis (i) first implies that the stable rank of $e_0^* H_0$ is $h_0 = \mathrm{rank}\ \mathrm{Fil}_0\, H_0$ (the slope-zero parts of the Newton and Hodge polygons of $e_0^* H$ coincide). Let s_0 be a point of $\mathrm{Spec}(A_0)$ with values in an algebraically closed extension of the fraction field of A_0. The proof of 1.2.4 shows that

$$\mathrm{rank\,.st}(s_0^* H_0) \geq \mathrm{rank\,.st}(e_0^* H_0)$$

(rank.st = stable rank), while (1.3.3.1) also gives

$$\mathrm{rank\,.st}(s_0^* H_0) \leq h_0.$$

Hence

$$\mathrm{rank\,.st}(e_0^* H_0) = \mathrm{rank\,.st}(s_0^* H_0) = h_0.$$

It follows, on the one hand, that

$$\mathrm{Fil}_0(s_0^* H_0) = (s_0^* H_0)^{s} s, \qquad \mathrm{Fil}^1(s_0^* H_0) = (s_0^* H_0)^{n}\mathrm{ilp},$$

so $s_0^* \mathrm{Fil}_0\, H_0 \cap s_0^* \mathrm{Fil}^1\, H_0 = 0$ and therefore

$$(1.3.3.2) \qquad H_0 = \mathrm{Fil}_0\, H_0 \oplus \mathrm{Fil}^1\, H_0.$$

On the other hand, by 1.2.4 there is an exact sequence of F-crystals over A_0,

$$(1.3.3.3) \qquad 0 \to U_0 \to H \to E \to 0,$$

where U_0 is the unit sub-F-crystal of H and E is a topologically nilpotent F-crystal. Since by definition $\mathrm{rank}(U_0) = \mathrm{rank\,.st}(e_0^* H_0) = h_0$, U_0 lifts $\mathrm{Fil}_0\, H_0$; hence the projection $H \to E$ induces an isomorphism $\mathrm{Fil}^1\, H_0 \simeq E_0$. This means that $F(\varphi) : \varphi^* E \to E$ is divisible by p, so $E = E'(-1)$, where E' is an F-crystal of level $\leq n-1$. Clearly $e_0^* E$, and hence also $e_0^* E'$, has the same Newton and Hodge polygons; thus, by the induction hypothesis applied to E', one

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concludes that the Hodge and conjugate filtrations of $E_0 = E \otimes A_0$ are opposite. Thus (ii) will be proved once we verify that the projection $H \to E$ induces isomorphisms, for $i \geq 0$,

$$\mathrm{Fil}_i H_0 / \mathrm{Fil}_0 H_0 \simeq \mathrm{Fil}_i E_0, \qquad \mathrm{Fil}^{i+1} H_0 \simeq \mathrm{Fil}^{i+1} E_0.$$

These maps are plainly injective, and their surjectivity follows easily from the fact that U_0 is a unit F-crystal. We now prove (ii) $\Rightarrow$ (iii). Again suppose that H has level $\leq n$ and that the conclusion is known for F-crystals of level $\leq n-1$. Uniqueness in (1.3.2.1) is clear by induction, since U_0 must be the unit sub-F-crystal of H. For existence, note that (1.3.3.2) holds by hypothesis. By (1.3.3.1),

$$e_0^* \mathrm{Fil}_0 H_0 = (e_0^* H_0)^s s, \qquad s_0^* \mathrm{Fil}_0 H_0 = (s_0^* H_0)^s s$$

(the decomposition into semisimple and nilpotent parts is unique), so the stable ranks of $e_0^* H_0$ and $s_0^* H_0$ are equal. By 1.2.4 we again obtain the exact sequence (1.3.3.3). As seen above, the Hodge and conjugate filtrations of E_0 are obtained from those of H_0 by passage to the quotient, so they are opposite. Since $E = E'(-1)$, with E' of level $\leq n-1$, induction gives a filtration of E by sub-F-crystals

$$0 = W_0 \subset W_1 \subset \dots \subset W_i \subset W_{i+1} \subset \dots$$

such that W_i lifts $\mathrm{Fil}_i E_0$ and W_i / W_{i-1} is of the form $T_i(-i)$, with T_i a unit F-crystal. For $i \geq 1$, let U_i be the inverse image of W_i in H. The U_i for $i \geq 1$, together with U_0, form a filtration (1.3.2.1) satisfying the stated conditions in (iii). This completes the proof of 1.3.2.

REMARK 1.3.4. If $A_0 = k$, the filtration (1.3.2.1) admits a unique splitting

$$(1.3.4.1) \qquad U_i = \oplus_{j \leq i} V_j(-j),$$

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where V_j is a unit F-crystal. This is a special case of Katz's theorem [10, 1.6.1], but it is easy to verify directly: uniqueness is clear, since for $j < i$ one has $\mathrm{Hom}(V_i(-i), V_j(-j)) = 0$; existence of the splitting follows readily by induction on the level, using a fixed-point argument analogous to those used in the proof of 1.2.4.

1.3.5. Let H be an F-crystal over A_0. A Hodge filtration on H means a finite decreasing filtration by free A-submodules

$$\mathrm{Fil}^0 H = H \supset \mathrm{Fil}^1 H \supset \ldots \supset \mathrm{Fil}^i H \supset \mathrm{Fil}^{i+1} H \supset \ldots$$

satisfying the following two conditions:

(i) for every i, $\mathrm{Fil}^i H$ lifts $\mathrm{Fil}^i H_0$;

(ii) ("transversality") for every i,

$$(1.3.5.1) \qquad \nabla \mathrm{Fil}^i H \subset \Omega^1_{A/W} \otimes \mathrm{Fil}^{i-1} H.$$

A Hodge F-crystal over A_0 is an F-crystal over A_0 equipped with a Hodge filtration.

A standard example, in the geometric situation considered in 1.3.1, is given by a lifting of X_0/A_0 to a proper smooth formal scheme X/A such that the Hodge spectral sequence of X/A,

$$E_1^{\mathrm{ij}} = H^j(X, \Omega^i_{X/A}) \Rightarrow H^*_{\mathrm{DR}}(X/A)\,(= H^*(X_0/A)),$$

degenerates at E_1, with E_1 free over A. The Hodge filtration on $H^n_{\mathrm{DR}}(X/A)$, as the abutment of this spectral sequence, satisfies conditions (i) and (ii) above.

PROPOSITION 1.3.6. Let $(H, \mathrm{Fil}^\bullet H)$ be a Hodge F-crystal over A_0. If H is ordinary (1.3.3), the filtrations U_i (1.3.2.1) and $\mathrm{Fil}^\bullet H$ are opposite; that is, for every i, $H = U_i \oplus \mathrm{Fil}^{i+1} H$. Hence there is a decomposition

$$(1.3.6.1) \qquad H = \oplus_i H^i, \qquad H^i = U_i \cap \mathrm{Fil}^i H,$$

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with H^i of rank h_i (1.3.1.5).
This is an immediate consequence of 1.3.2 and 1.3.5(i) (condition (ii) is not used).
In the preceding example, suppose the F-crystal $H = H^n_{\mathrm{DR}}(X/A)$ is ordinary. For every $r \geq 1$, the conjugate spectral sequence

$$(*)_r \qquad E_2^{\mathrm{ij}} = H^i(X \otimes W_r, H^j(\Omega^\bullet{}_{X\otimes W_r/A\otimes W_r})) \Rightarrow H^*_{\mathrm{DR}}(X \otimes W_r/A \otimes W_r)$$

defines a filtration $(U_{i,r})$ on $H^n_{\mathrm{DR}}(X \otimes W_r/A \otimes W_r)$. One may hope that the projective limit of the spectral sequences $(*)_r$ is a spectral sequence whose abutment filtration on H^n is the limit of the $(U_{i,r})$ and coincides with the filtration (U_i).

1.4. Ordinary Hodge F-crystals of level ≤ 1.

1.4.1. Let H be an ordinary Hodge F-crystal of level $\leq$ 1 over A_0. By 1.3.2, there is an extension of F-crystals

$$(1.4.1.1) \qquad 0 \to U \to H \to V(-1) \to 0,$$

where U and V are unit F-crystals and U lifts $\mathrm{Fil}_0\, H_0$.
Let r (respectively s) be the rank of U (respectively V). The module H also has the free submodule $\mathrm{Fil}^1\, H$, which lifts $\mathrm{Fil}^1\, H_0$ and therefore satisfies, by (1.3.1.4),

$$(1.4.1.2) \qquad F(\varphi)\varphi^*\,\mathrm{Fil}^1\, H \subset \mathrm{pH}$$

for every lifting φ of Frobenius. Moreover, by 1.3.6, H decomposes as an A-module direct sum

$$(1.4.1.3) \qquad H = U \oplus \mathrm{Fil}^1\, H.$$

In particular, $\mathrm{Fil}^1\, H$ has rank s and maps isomorphically onto V.
THEOREM 1.4.2. With the notation of 1.4.1: (i) There exist a basis $\mathrm{a} = (a_i)_{1 \leq i \leq r}$ of the A-module U and a basis $b = (b_i)_{1 \leq i \leq s}$ of the A-module $\mathrm{Fil}^1\, H$ satisfying the conditions

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$$(1.4.2.1) \qquad \nabla a_i = 0, \qquad 1 \le i \le r,$$

$$(1.4.2.2) \qquad \nabla b_i = \sum_{1 \le j \le r} \eta_{ij} \otimes a_j, \qquad \eta_{ij} \in \Omega^1_{A/W}, \qquad 1 \le i \le s,$$

$$(1.4.2.3) \qquad F(\varphi)\varphi^* a_i = a_i, \qquad 1 \le i \le r,$$

$$(1.4.2.4) \qquad F(\varphi)\varphi^* b_i = p\, b_i + p \sum_{1 \le j \le r} u_{ij}(\varphi) a_j, \qquad u_{ij}(\varphi) \in A, \qquad 1 \le i \le s,$$

for every lifting φ of Frobenius. Moreover, the forms η_{ij} are closed and satisfy, for every lifting φ of Frobenius,

$$(1.4.2.5) \qquad \varphi^* \eta_{ij} = p\eta_{ij} + p\, \mathrm{d}u_{ij}(\varphi).$$

(ii) Once the bases a and b have been chosen as in (i), there is a unique family of formal power series $\tau_{ij} \in$ K[[t]] (where K is the fraction field of W) such that, for all i and j,

$$(1.4.2.6) \qquad \eta_{ij} = d\tau_{ij},$$

$$(1.4.2.7) \qquad \varphi^* \tau_{ij} - p\tau_{ij} - \mathrm{pu}_{ij}(\varphi) = 0$$

for every lifting φ of Frobenius. One has

$$(1.4.2.8) \qquad \tau_{ij}(0) \in \mathrm{pW}.$$

Moreover, if $p \neq 2$, the series

$$(1.4.2.9) \qquad q_{ij} = \exp(\tau_{ij})$$

are defined, belong to A, and satisfy $q_{ij}(0) = 1$ mod pW.

We prove (i). By the structure theorem for unit F-crystals (1.2.2), there is a basis a = $(a_i)_{1 \le i \le r}$ of U satisfying (1.4.2.1) and (1.4.2.3), and a basis $b = (b_i)_{1 \le i \le s}$ of $\mathrm{Fil}^1 H$ such that, if b'_i denotes the image of b_i in $V(-1)$, then $\nabla b'_i = 0$ and $F(\varphi)\varphi^* b'_i = p\, b'_i$ for every i and every lifting φ of Frobenius. Hence ∇b_i and $F(\varphi)\varphi^* b_i$ have the forms (1.4.2.2) and (1.4.2.4). Applying ∇ to (1.4.2.2), integrability gives $d\eta_{ij} = 0$ for all i, j. Applying ∇ to (1.4.2.4) and using horizontality of F (1.1.3.3) gives the relations (1.4.2.5).

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We prove (ii). Since the forms η_{ij} are closed, the Poincaré lemma gives series $\tau_{\mathrm{ij}} \in$ K[[t]], determined up to an additive constant, such that (1.4.2.6) holds. Relation (1.4.2.5) then implies

$$\varphi^*\tau_{\mathrm{ij}} - p\tau_{\mathrm{ij}} - \mathrm{pu}_{\mathrm{ij}}(\varphi) \in K,$$

and hence

$$(1.4.2.10) \qquad \varphi^*\tau_{\mathrm{ij}} - p\tau_{\mathrm{ij}} - \mathrm{pu}_{\mathrm{ij}}(\varphi) = (\varphi^*\tau_{\mathrm{ij}})(0) - p\tau_{\mathrm{ij}}(0) - \mathrm{pu}_{\mathrm{ij}}(\varphi)(0).$$

Write $x \mapsto x^{\sigma}$ for the Frobenius automorphism of K. First fix the lifting φ of Frobenius given by $\varphi(t_i) = t_i^p$, and normalize τ_{ij} by the condition

$$(1.4.2.11) \qquad (\varphi^*\tau_{\mathrm{ij}})(0) - p\tau_{\mathrm{ij}}(0) - \mathrm{pu}_{\mathrm{ij}}(\varphi)(0) = 0,$$

which may also be written, since $(\varphi^*\tau_{\mathrm{ij}})(0) = \tau_{\mathrm{ij}}(0)^{\sigma}$,

$$(1.4.2.12) \qquad \tau_{\mathrm{ij}}(0) = \sum\nolimits_{n\geq 1} V^n u_{\mathrm{ij}}(\varphi)(0)$$

(where $\mathrm{Vx} = \mathrm{pF}^{-1}x = \mathrm{px}^{\sigma^{-1}}$). In particular, (1.4.2.8) holds. We show that if ψ is any lifting of Frobenius, then (1.4.2.7), with φ replaced by ψ, is satisfied. In view of (1.4.2.10), again with φ replaced by ψ, it is equivalent to verify

$$(1.4.2.13) \qquad (\psi^*\tau_{\mathrm{ij}})(0) - p\tau_{\mathrm{ij}}(0) - \mathrm{pu}_{\mathrm{ij}}(\psi)(0) = 0.$$

If $\psi(0) = 0$ (that is, ψ is compatible with the augmentation $e :$ W[[t]] $\to W$), then $(\psi^*\tau_{\mathrm{ij}})(0) = \tau_{\mathrm{ij}}(0)^{\sigma}$, and (1.4.2.13) is equivalent to (1.4.2.12) with φ replaced by ψ. It therefore remains only to check that $u_{\mathrm{ij}}(\varphi)(0) = u_{\mathrm{ij}}(\psi)(0)$. The augmentation e is the Teichmüller lifting (1.1.4) of k[[t]] $\to k$ simultaneously for φ and ψ; hence $F(\varphi)\varphi^*$ and $F(\psi)\psi^*$ induce the same $\sigma-$linear endomorphism F of e^*H. Applying e^* to (1.4.2.4), written once for φ and once for ψ, gives

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$$F(e^*b_i) = p(e^*b_i) + p\sum_{1\le j\le r} u_{ij}(\varphi)(0)(e^*a_j)$$
$$= p(e^*b_i) + p\sum_{1\le j\le r} u_{ij}(\psi)(0)(e^*a_j),$$

so $u_{ij}(\varphi)(0) = u_{ij}(\psi)(0)$, proving (1.4.2.13) in this case.
Now let ψ be an arbitrary lifting of Frobenius. We may write

$$\psi(t_i) = \psi_0(t_i) + \mathrm{pw}_i, \qquad 1 \le i \le n,$$

with $w = (w_1, \dots, w_n) \in W^n$, where $\psi_0 : A \to A$ is a lifting of Frobenius such that $\psi_0(0) = 0$. Applying (1.1.3.4), with $\varphi = \psi_0$ and $x = b_i$, and using (1.4.2.4), gives

$$(*) \qquad \sum_{1\le j\le r} \mathrm{pu}_{ij}(\psi)a_j = \sum_{1\le j\le r} \mathrm{pu}_{ij}(\psi_0)a_j$$
$$+ \sum_{|n|>0} p^{[n]}(w^\sigma)^n F(\psi_0)\psi_0^*(\nabla(D)^n b_i).$$

Let us compute $\nabla(D)^n b_i$. If $D_k = \partial/\partial t_k$, then by (1.4.2.2) and (1.4.2.6),

$$\nabla(D_k)b_i = \langle D_k, \sum_j d\tau_{ij} \otimes a_j\rangle = \sum_j (D_k\tau_{ij})a_j.$$

Since $\nabla a_j = 0$, it follows that

$$\nabla(D^n)b_i = \sum_j (D^n\tau_{ij})a_j$$

for every multi-index n with |n| > 0. Substitution in (*) yields
$\mathrm{pu}_{ij}(\psi) = \mathrm{pu}_{ij}(\psi_0) + \sum_{|n|>0} p^{[n]}(w^\sigma)^n \psi_0^*(D^n\tau_{ij})$,
and the identity (1.4.2.13) to be proved becomes

$$(**) \qquad (\psi^*\tau_{ij})(0) - p\tau_{ij}(0) - \mathrm{pu}_{ij}(\psi_0)(0)$$
$$- \sum_{|n|>0} p^{[n]}(w^n(D^n\tau_{ij})(0))^\sigma = 0.$$

Write $\psi = \psi_0 \circ$ a, where a is the W-algebra endomorphism of A given by $a(t_i) = t_i + \mathrm{pw}_i$. Then

$$(\psi^*\tau_{ij})(0) = (\psi_0^*(a^*\tau_{ij}))(0) = (a^*\tau_{ij})(0)^\sigma = \tau_{ij}(\mathrm{pw})^\sigma.$$

Moreover ψ_0 satisfies (1.4.2.13), as shown above, because $\psi_0(0) = 0$ and $(\psi_0^*\tau_{ij})(0) = \tau_{ij}(0)^\sigma$. Thus

$$p\tau_{ij}(0) = \tau_{ij}(0)^\sigma - \mathrm{pu}_{ij}(\psi_0)(0).$$

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We may therefore rewrite (**) as

$$\tau_{ij}(pw)^\sigma = \tau_{ij}(0)^\sigma + \sum_{|n|>0} p^{[n]}(w^n(D^n\tau_{ij})(0))^\sigma,$$

i.e.

$$\tau_{ij}(pw) = \tau_{ij}(0) + \sum_{|n|>0} p^{[n]} w^n (D^n \tau_{ij})(0).$$

This is precisely the Taylor expansion of $\tau_{ij}(pw)$. It proves (**), and therefore (1.4.2.13), in every case. To prove the last assertion of 1.4.2 we shall use the following result of Dwork:

LEMMA 1.4.3 ([5], Lemma 1) (*). Let φ denote the $\sigma-$linear automorphism of K[[t]] (where $t = (t_1, \ldots, t_n)$ and K is the fraction field of W) given by $t_i \mapsto t_i^p$. Let $f \in$ K[[t]] with f(0) = 1. The following conditions are equivalent:

$$(i)\ f \in W[[t]];$$

(ii) $(\varphi^* f)/f^p = 1 + pu$, with $u \in$ W[[t]] and u(0) = 0.

The implication $(i) \Rightarrow$ (ii) is immediate. Recall the proof of (ii) $\Rightarrow$ (i). Write $f = \sum_i a_i t^i$ $(a_0 = 1)$, and suppose it is known that $a_i \in W$ for |i| < r, where |i| = $i_1 + \ldots + i_n$. Denote by ( )<j the part of total degree < j. Then

$((\sum_{|i|\geq 0} a_i t^i)^p) < r$
$= ((\sum_{|i|\leq r-1} a_i t^i)^p) < r + p\sum_{|i|=r} a_i t^i$
$= (\sum_{|i|\leq r-1} a_i^p t^{\mathrm{pi}}) < r + p\sum_{|i|=r} a_i t^i$ mod pW[[t]].

Next, writing $f^p/(\varphi^* f) = 1 + p\sum_{|i|\geq 1} d_i t^i$ with $d_i \in W$, one obtains

$((\sum_{|i|\geq 0} a_i^\sigma t^{\mathrm{pi}})(1 + p\sum_{|i|\geq 1} d_i t^i)) < r$
$= ((\sum_{|i|\leq r-1} a_i^\sigma t^{\mathrm{pi}})(1 + p\sum_{|i|\geq 1} d_i t^i)) < r$
$= (\sum_{|i|\leq r-1} a_i^\sigma t^{\mathrm{pi}}) < r$ mod pW[[t]]
$= (\sum_{|i|\leq r-1} a_i^p t^{\mathrm{pi}}) < r$ mod pW[[t]].

———————

(*) See also Hazewinkel [7] for a generalization.

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Comparing the two expressions obtained for $(f^p) < r$ gives

$$p \sum_{|i|=r} a_i t^i \in \mathrm{pW}[[t]],$$

so $a_i \in W$ for every i with |i| = r. This proves (ii) $\Rightarrow$ (i).

Before returning to the proof of 1.4.2, we record two immediate consequences of 1.4.3.

COROLLARY 1.4.4. Let $g \in$ K[[t]] with g(0) = 0. The following conditions are equivalent:

$$(i)\ \exp(g) \in W[[t]];$$

(ii) $\varphi^* g -$ pg $\in$ pW[[t]].

Indeed, put f = exp(g). Since for $n \geq 1$ the quantities $p^n/n!$ (and a fortiori p^n/n) belong to pW, condition (ii) is equivalent to $\varphi^* f/f^p$ = 1 + pu, with $u \in$ W[[t]] and u(0) = 0.

COROLLARY 1.4.5. Suppose $p \neq 2$. Let $g \in$ K[[t]] satisfy g(0) $\in$ pW and $\varphi^* g -$ pg $\in$ pW[[t]]. Then the series f = exp(g) is defined, belongs to W[[t]], and satisfies f(0) = 1 mod p.

Indeed, let $h = g -$ g(0). Since $p \neq 2$ and g(0) $\in$ pW, exp(g(0)) is defined, and hence so is exp(g) = exp(g(0))exp(h). Write g(0) = a, with a $\in W$. Then

$$\varphi^* h - \mathrm{ph} = \varphi^* g - \mathrm{pg} - (a^\sigma - \mathrm{pa}) \in \mathrm{pW}[[t]]$$

because $a^\sigma -$ pa $\in$ pW and $\varphi^* g -$ pg $\in$ pW[[t]]. Applying 1.4.4 to h gives exp(h) $\in$ W[[t]], and since exp(a) = 1 mod pW, it follows that f = exp(g) $\in$ W[[t]] and f(0) = 1 mod pW.

End of the proof of 1.4.2. It remains only to apply 1.4.5 to τ_{ij}: its hypotheses hold by (1.4.2.7) and (1.4.2.8).

1.4.6. If H is a Hodge F-crystal as in 1.3.5, the connection ∇ induces, by (1.3.5.1), an A-linear homomorphism

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$$(1.4.6.1) \qquad \mathrm{gr}\,\nabla : \mathrm{gr}^i H \to \Omega^1_{A/W} \otimes \mathrm{gr}^{i-1} H$$

(where $\mathrm{gr}^i = \mathrm{Fil}^i / \mathrm{Fil}^{i+1}$). In particular, in the situation of 1.4.1, ∇ induces an A-linear homomorphism

$$(1.4.6.2) \qquad \mathrm{gr}\,\nabla : \mathrm{Fil}^1 H \to \Omega^1_{A/W} \otimes U.$$

Since $U \simeq H / \mathrm{Fil}^1 H$ by (1.4.1.3), this gives an A-linear homomorphism

$$(1.4.6.3) \qquad \mathrm{gr}\,\nabla : T_{A/W} \to \mathrm{Hom}_A(\mathrm{Fil}^1 H, U),$$

where $T_{A/W} = (\Omega^1_{A/W}) \vee$.
COROLLARY 1.4.7. Under the hypotheses of 1.4.1, suppose that (1.4.6.3) is an isomorphism and that p is different from 2. Then the elements $q_{\mathrm{ij}} - 1$ ($1 \leq i \leq s$, $1 \leq j \leq r$) defined in (1.4.2.9), together with p, form a regular system of parameters for A = W[[t]]; that is, the W-homomorphism

$$W[[x_{\mathrm{ij}}]]_{(1 \leq i \leq s, 1 \leq j \leq r)} \to A$$

sending x_{ij} to $q_{\mathrm{ij}} - 1$ is an isomorphism. If φ denotes the lifting of Frobenius defined by $\varphi(q_{\mathrm{ij}}) = q_{\mathrm{ij}}^p$, then

$$(1.4.7.1) \qquad F(\varphi)\varphi^* b_i = p\, b_i$$

for every i, with the notation of 1.4.2.
Let $m = (p, t_1, \dots, t_n)$ be the maximal ideal of A = W[[t]]. Since $q_{\mathrm{ij}}(0) = 1$ mod pW, the q_{ij} are units of A congruent to 1 mod m. To prove the first assertion it is enough to show that the forms $\eta_{\mathrm{ij}} = d$ log q_{ij} form an A-basis of $\Omega^1_{A/W}$.
Every homomorphism $f : \mathrm{Fil}^1 H \to U$ is determined by a matrix (f_{ij}) such that $f(b_i) = \sum_j f_{\mathrm{ij}} a_j$. Let D_{ij} be the basis of $T_{A/W}$ characterized by

$$(\mathrm{gr}\,\nabla)(D_{\mathrm{ij}})_k \ell = \{ 0 \text{ if } (k, \ell) \neq (i, j),$$
$$1 \text{ if } (k, \ell) = (i, j). \}$$

For every $D \in T_{A/W}$, (1.4.2.2) gives
$(\mathrm{gr}\ \nabla)(D) b_i = \sum_j \eta_{\mathrm{ij}}(D) a_j$,

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thus
$(\mathrm{gr}\ \nabla)(D_{\mathrm{ij}})_k \ell = \eta_k \ell(D_{\mathrm{ij}})$.

The η_{ij} therefore form a basis of $\Omega^1_{A/W}$, namely the basis dual to the basis D_{ij}. On the other hand, the relation $\varphi^* q_{\mathrm{ij}} = q_{\mathrm{ij}}^p$ implies $\varphi^* \log q_{\mathrm{ij}} = p \log q_{\mathrm{ij}}$; but since $\log q_{\mathrm{ij}} = \tau_{\mathrm{ij}}$, (1.4.2.7) implies $u_{\mathrm{ij}}(\varphi) = 0$, and hence (1.4.7.1) follows from (1.4.2.4).

1.4.8. Let, as in 1.3.6, $(H, \mathrm{Fil}\bullet H)$ be a Hodge F-crystal over A_0 (= k[[t]]) such that H is ordinary. Suppose that H has level $\leq N$, i.e. (1.3.1) $\mathrm{Fil}^i H_0 = 0$ for $i \geq$ n+1, so that the decomposition (1.3.6.1) is written

$$(1.4.8.1) \qquad H = \oplus_{0\leq i\leq N} H^i, \qquad H^i = U_i \cap \mathrm{Fil}^i H, \qquad \mathrm{rg}(H^i) = h_i.$$

Suppose moreover that $p \neq 2$. Applying 1.4.2 to the ordinary Hodge F-crystals (of level ≤ 1)
$(U_{i+2}/U_i, (\mathrm{Fil}\bullet H \cap U_{i+2})/(\mathrm{Fil}\bullet H \cap U_i))$,
one obtains:

COROLLARY 1.4.9. Under the hypotheses of 1.4.8, there exist a basis $(e^i_1, \dots, e^i_{h_i})_{0\leq i\leq N}$ of H and elements $q_{i;\alpha,\beta}$ of A ($i \leq N-1$, $\alpha \leq h_{i+1}$, $\beta \leq h_i$) such that

$$(1.4.9.1) \qquad \{ q_{i;\alpha,\beta}(0) \in 1 + \mathrm{pW},$$
$$\nabla e^0_i = 0 \qquad (1 \leq i \leq h_0),$$
$$\nabla e^m_i = \sum_{1\leq j\leq h_{m-1}} (d \log q_{m-1;i,j}) e^{m-1}_j \qquad (1 \leq m \leq N, 0 \leq i \leq h_m).\}$$

One may use 1.4.9 to bound the growth of the horizontal sections of H. Recall [8, 3.1] that these “converge in the open unit polydisc,” and, more precisely, that the module $(H \otimes_A K\{\{t_1, \dots, t_n\}\}, \nabla)$ possesses a basis of horizontal sections; this is a property valid for every F-crystal, independently of any ordinarity hypothesis. That said, it follows readily from 1.4.9 that, for every i, the horizontal sections of $U_i \otimes_A$ K{{t}} belong to $U_i \otimes_A L_i$, where

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$$0 = L_{-1} \subset L_0 \subset \ldots \subset L_i \subset \ldots$$

is the sequence of A-submodules of K{{t}} defined recursively by

$$\begin{aligned}&(f \in L_i) \Leftrightarrow (f \text{ belongs to the } A-\text{submodule of } K\{\{t\}\} \text{ generated}\\&\text{by the series } g \text{ such that } \mathrm{dg} = \sum a_\alpha\, d \log u_\alpha,\\&\text{with } a_\alpha \in L_{i-1} \text{ and } u_\alpha \in A,\ u_\alpha(0) \in 1 + \mathrm{pW}).\end{aligned}$$

In the situation of 1.4.8, however, one cannot expect a statement analogous to 1.4.7 (the hypothesis that (1.4.6.3) is an isomorphism has no generalization in level ≥ 2).

2. GEOMETRIC APPLICATIONS

We retain the notation A_0, A of 1.1.1.

2.1. Canonical coordinates.

A) Abelian varieties (cf. [8, §8]).

2.1.1. Let X/A be a formal abelian scheme of relative dimension g. The A-module

$$H = H^1_{\mathrm{DR}}(X/A),$$

endowed with the Gauss-Manin connection ∇, is an F-crystal over A_0 of rank 2g. It is known (see, for example, [13, Addendum]) that the Hodge spectral sequence

$$E_1^{\mathrm{ij}} = H^j(X, \Omega^i_{X/A}) \Rightarrow H^*_{\mathrm{DR}}(X/A)$$

degenerates at E_1, and that the term E_1^{ij} is free over A and is compatible with every base change. It follows from (1.3.5) that H, endowed with the Hodge filtration

$$H = \mathrm{Fil}^0 H \supset \mathrm{Fil}^1 H\, (= H^0(X, \Omega^1_{X/A})) \supset 0,$$

is a Hodge F-crystal over A_0. Note also that, if $X_0 = X \otimes A_0$, the Hodge spectral sequence and the conjugate spectral sequence of X_0/A_0 degenerate (at E_1 and E_2, respectively), and that their initial terms are

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free and compatible with every base change. In particular, the associated graded object for the conjugate filtration, $\mathrm{gr}_{\bullet}(H \otimes A_0)$, is free. On the other hand, if $e_0 : A_0 \to k$ is the augmentation, the F-crystal $e_0^* H$ induced over k (1.1.4) is precisely $H^1(X \otimes k/W)$, because, if $e : A \to W$ is the augmentation, then $e^* H = H^1_{\mathrm{DR}}(X \otimes W/W) = H^1(X \otimes k/W)$ by the definition of crystalline cohomology.

2.1.2. Suppose that the abelian variety $X \otimes k$ is ordinary, which means, equivalently, that $(X \otimes \mathrm{k})(\mathrm{k}) \simeq (\mathbb{Z}/p)^g$, that Frobenius on $H^1(X \otimes k, \mathcal{O})$ is bijective, or that $H^1(X \otimes k/W)$ has the same Newton and Hodge polygons. From what has just been recalled, the F-crystal H is then ordinary in the sense of 1.3.3; and since it is a Hodge F-crystal of level ≤ 1, 1.4.2 applies to it. In the decomposition $H = U \oplus \mathrm{Fil}^1 H$, both U and $\mathrm{Fil}^1 H$ are free of rank g. There therefore exist a basis $(a_i)_{1 \leq i \leq g}$ of U, a basis $(b_i)_{1 \leq i \leq g}$ of $\mathrm{Fil}^1 H$, and series $\tau_{\mathrm{ij}} \in \mathrm{K}[[\mathrm{t}]]$ $(1 \leq j \leq g, 1 \leq j \leq g)$ satisfying conditions (1.4.2.1) through (1.4.2.8). Moreover, if $p \neq 2$, one has the series $q_{\mathrm{ij}} = \exp(\tau_{\mathrm{ij}}) \in A$, with $q_{\mathrm{ij}}(0) = 1$ mod pW.

2.1.3. Suppose that X/A is the versal formal deformation of $X \otimes k$, so that $A = W[[t_1, \ldots, t_{g^2}]]$. Then (1.4.6.3) is an isomorphism, identified with the Kodaira-Spencer isomorphism

$$T_{A/W} \simeq H^1(X, T_{X/A}) \simeq \mathrm{Hom}(H^0(X, \Omega^1_{X/A}), H^1(X, \mathcal{O}))$$

(cf. (IV 2.3)). Thus, if $p \neq 2$, the elements $q_{\mathrm{ij}} - 1 \in A$ form, by 1.4.7, “canonical” coordinates on A (i.e. $A \simeq W[[q_{\mathrm{ij}} - 1]]$); and if φ denotes the lifting of Frobenius given by $\varphi(q_{\mathrm{ij}}) = q_{\mathrm{ij}}^p$, then, with the notation of 2.1.2,

$$F(\varphi)\varphi^* a_i = a_i, \qquad F(\varphi)\varphi^* b_i = p\, b_i \qquad (1 \leq i \leq g).$$

Let $e : A \to W$ be the Teichmüller lifting of e_0 corresponding to φ, i.e. such that $e(q_{\mathrm{ij}}) = 1$. It is not difficult to show - we

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shall establish an analogous result below for K3 surfaces - that e is independent of the choice of (a, b, q), and that there is a canonical isomorphism (defined using (a, b, q), but independent of it) between S and G_W, where G is the formal torus over $\mathbb{Z}_p$ with character group $\mathrm{Hom}_{\mathbb{Z}_p}(P_0, P_1)$, P_0 (respectively P_1) denoting the $\mathbb{Z}_p -$submodule (free of rank g) of $H^1(X \otimes k/W)$ consisting of the x such that Fx = x (respectively Fx = px). This isomorphism sends e to the identity element of G_W. One can show (see the Appendix and the following Exposé, by N. Katz) that this formal group structure on S coincides with the one defined, in the manner of Serre-Tate, by the versal formal lifting of the p-divisible group associated with $X \otimes k$. In particular, the abelian variety $e^* X/W$ is the canonical lifting of ($X \otimes$ k)/k [14, V §3]. Recall that this lifting corresponds to the trivial lifting (i.e. as a product) of the p-divisible group associated with $X \otimes k$, or, equivalently, to the lifting of $\mathrm{Fil}^1_{\mathrm{Hdg}}\ H^1_{\mathrm{DR}}(X \otimes k/k)$ to the direct factor $P_1 \otimes W$ of slope 1 in $H^1(X \otimes k/W)$ (cf. 1.3.4).

2.1.4. Variant. Let X/A be a proper smooth curve with geometrically connected fibers of genus $g \geq 1$. Then $H = H^1_{\mathrm{DR}}(X/A)$, endowed with the Gauss-Manin connection ∇ and the Hodge filtration $\mathrm{Fil}^1\ H = H^0(X, \Omega^1_{X/A})$, is a Hodge F-crystal of level 1. If $X \otimes k$ is assumed ordinary, 1.4.2 applies and gives the results of Katz [8, §8]: if $(a_i)_{1 \leq i \leq g}$ is a basis of horizontal sections fixed by F in the unit subcrystal U, then one may take as the basis $(b_i)_{1 \leq i \leq g}$ of $\mathrm{Fil}^1\ H$ the basis dual to (a_i) under Poincaré duality ((a, b) is then a symplectic basis of H), because compatibility of the duality with F and ∇ immediately shows that the conditions of (1.2.4 (i)) are satisfied.

B) K3 surfaces.

2.1.5. Let X/A be a proper smooth formal scheme such that $X_k = X \otimes k$ is a K3 surface. Then, by (IV 2.2, 2.3), the

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A-module

$$H = H^2_{\mathrm{DR}}(X/A),$$

endowed with the Gauss-Manin connection ∇ and the Hodge filtration

$$H = \mathrm{Fil}^0 H \supset \mathrm{Fil}^1 H \supset \mathrm{Fil}^2 H \supset 0,$$

is a Hodge F-crystal over A_0. The submodules $\mathrm{Fil}^i H$ are free of ranks 22, 21, and 1 for $i = 0,1,2$, and the associated graded object $\mathrm{gr}^\bullet H$ is free. Moreover, the cup product

$$\langle , \rangle : H \otimes H \to H^4_{\mathrm{DR}}(X/A) \simeq A$$

is a perfect duality, horizontal for ∇, i.e. satisfying

$$\langle \nabla x, y\rangle + \langle x, \nabla y\rangle = \nabla\langle x, y\rangle,$$

and compatible with F, i.e. such that, for every lifting of Frobenius φ,

$$\langle F(\varphi)\varphi^* x, F(\varphi)\varphi^* y\rangle = F(\varphi)\varphi^*\langle x, y\rangle.$$

Observe that $F(\varphi)\varphi^*|_{H^4_{\mathrm{DR}}(X/A)} = p^2\alpha$, where α is an automorphism; that is, $H^4_{\mathrm{DR}}(X/A)(2)$ is a unit F-crystal (of rank 1), which we identify with the trivial F-crystal A by means of a horizontal basis fixed by F. Recall also that

$$\mathrm{Fil}^1 H = (\mathrm{Fil}^2 H)\perp .$$

Finally, recall that, if $e_0 : A_0 \to k$ is the augmentation, the F-crystal $e_0^* H$ (1.1.4) is precisely the crystalline cohomology $H^2(X_k/W)$.

2.1.6. Suppose now that X_k is ordinary, which means, by definition, that the Newton polygon of $H^2(X_k/W)$ coincides with its Hodge polygon, i.e. has slopes (0,1,2) with multiplicities (1,20,1). Equivalently, Frobenius on $H^2(X_k, \mathcal{O})$ is nonzero. Since $e_0^* H = H^2(X_k/W)$ and $\mathrm{gr}^\bullet H$ is free, the F-crystal H is then ordinary in the sense of 1.3.3, and consequently admits a filtration by sub-F-crystals

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$$0 \subset U_0 \subset U_1 \subset U_2 = H$$

such that U_i/U_{i-1} is of the form $V_i(-i)$, where V_i is a unit F-crystal of rank 1, 20, and 1 for $i = 0,1,2$. Moreover, since U_i lifts $\mathrm{Fil}_i\, H_0$, compatibility of F with the duality implies

$$U_1 = U_0 \perp .$$

Finally, by (1.3.6), the filtrations (U_i) and $(\mathrm{Fil}^i\, H)$ are opposite; that is,

$$H = H_0 \oplus H_1 \oplus H_2,$$

with

$$H_0 = U_0, \qquad H_1 = U_1 \cap \mathrm{Fil}^1 H, \qquad H_2 = \mathrm{Fil}^2 H, \qquad H_1 \simeq U_1/U_0.$$

THEOREM 2.1.7. Assume $p \neq 2$. Let $X/A = W[[t_1, \dots, t_{20}]]$ be the universal formal deformation of an ordinary K3 surface X_k/k (IV 1.3). Then, with the notation of 2.1.5 and 2.1.6, there exist a basis $(a, b_1, \dots, b_{20}, c)$ of H adapted to the decomposition $H = H_0 \oplus H_1 \oplus H_2$ and satisfying $\langle a, c\rangle = 1$, and elements $q_i \in A$ $(1 \leq i \leq 20)$ having the following properties:

(i) $q_i(0) = 1 \mod \mathrm{pW}$, and the q_i define an isomorphism

$$A \simeq W[[q_1 - 1, \dots, q_{20} - 1]]$$

(i.e. the forms $d \log q_i$ form a basis of $\Omega^1_{A/W}$).

$$\text{(ii)} \begin{cases} \nabla a = 0, \\ \nabla b_i = (d \log q_i) a \qquad (1 \leq i \leq 20), \\ \nabla c = -\sum (d \log q_i) b_i \vee , \end{cases}$$

where $(b_i \vee)$ denotes the basis of H_1 dual to (b_i) for the restriction to H_1 of the cup-product form.
(iii) If $\varphi : A \to A$ denotes the lifting of Frobenius such that $\varphi(q_i) = q_i^p$, then

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$\{ F(\varphi)\varphi^* a = $ a,
$F(\varphi)\varphi^* b_i = p\, b_i$ $(1 \leq i \leq 20)$,
$F(\varphi)\varphi^* c = p^2 c.$ }

By 2.1.6, the F-crystal U_1, endowed with the filtration $\mathrm{Fil}^i\, H \cap U_1$, is an ordinary Hodge F-crystal of level 1. Moreover, since X/A is the universal formal deformation of X_k, the map

$\mathrm{gr}^1 \nabla : T_{A/W} \to \mathrm{Hom}_A(H_1, H_0) = \mathrm{Hom}_A(\mathrm{gr}^1 H, \mathrm{gr}^0 H)$

is an isomorphism by (IV 2.4). Applying 1.4.2 and 1.4.7 to U_1, there therefore exist a basis $(a, b_1, \ldots, b_{20})$ of U_1 adapted to the decomposition $U_1 = H_0 \oplus H_1$ and elements $q_i \in A$ satisfying conditions (i) through (iii) of 2.1.7. It remains to show that, if the basis c of H_2 is chosen so that $\langle a, c\rangle = 1$, the last formulas in (ii) and (iii) are satisfied. First, by the transversality condition (1.3.5.1), one has

$$\nabla c = \alpha \otimes c + \sum_{1 \leq i \leq 20} \beta_i \otimes b_i \vee .$$

The relation

$$0 = \nabla\langle a, c\rangle = \langle \nabla a, c\rangle + \langle a, \nabla c\rangle = \langle a, \nabla c\rangle$$

implies $\alpha = 0$. Then, from

$$0 = \nabla\langle b_i, c\rangle = \langle \nabla b_i, c\rangle + \langle b_i, \nabla c\rangle$$

and the second formula in (ii), one obtains $\beta_i = -\, d \log q_i$, which gives the last formula in (ii). On the other hand, from $F(\varphi)\varphi^* a =$ a and the relation

$$\langle F(\varphi)\varphi^* a, F(\varphi)\varphi^* c\rangle = p^2\langle a, c\rangle = p^2,$$

one obtains $F(\varphi)\varphi^* c = p^2 c$, completing the proof of 2.1.7.

2.1.8. Let X_k/k be an ordinary K3 surface. Set

$$(2.1.8.1) \qquad P_0 = H^2(X_k/W)_{F=1}, \qquad P_1 = H^2(X_k/W)_{F=p}, \qquad P_2 = H^2(X_k/W)_{F=p^2},$$

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where the notation $M_{F=a}$ denotes $\{x \in M \mid Fx = ax\}$. By 1.3.4, P_i is a free $\mathbb{Z}_p$ − module of rank 1, 20, and 1 for $i = 0,1,2$, and the F-crystal $H^2(X_k/W)$ has a unique decomposition

$$(2.1.8.2) \qquad H^2(X_k/W) = ((W,\sigma)\otimes P_0) \oplus ((W,p\sigma)\otimes P_1) \oplus ((W,p^2\sigma)\otimes P_2).$$

Suppose moreover that $p \neq 2$. With the notation of 2.1.7, let

$$(2.1.8.3) \qquad e_q : A \to W$$

denote the Teichmüller lifting of the augmentation $e_0 : A_0 \to k$ corresponding to the lifting φ considered in (iii), and hence given by $e_q(q_i) = 1$. Then X induces a formal scheme $e_q^* X/W$, and

$$(2.1.8.4) \qquad e_q^* H = H^2_{\mathrm{DR}}(e_q^* X/W) = H^2(X_k/W).$$

Moreover, by (2.1.7 (iii)), $e_q^* a$ is a basis of P_0, $(e_q^* b_i)_{1\le i\le 20}$ is a basis of P_1, $e_q^* c$ is a basis of P_2, and $\langle e_q^* a, e_q^* c\rangle = 1$. In particular, the Hodge filtration on $H^2_{\mathrm{DR}}(e_q^* X/W)$ is given by

$$(2.1.8.5) \qquad \mathrm{Fil}^1 H^2_{\mathrm{DR}}(e_q^* X/W) = (W\otimes P_1)\oplus(W\otimes P_2), \qquad \mathrm{Fil}^2 H^2_{\mathrm{DR}}(e_q^* X/W) = W \otimes P_2.$$

We shall deduce from this:

PROPOSITION 2.1.9. Under the hypotheses of 2.1.7, the lifting e_q (2.1.8.3) is independent of the family $q = (q_i)$ defined in 2.1.7.

We shall therefore write e in place of e_q. By analogy with the theory of the canonical lifting of ordinary abelian varieties [14, V §3], the formal scheme $e^* X/W$, which we shall denote simply by X_W, will be called the canonical lifting of X_k.

2.1.10. To prove 2.1.9, we shall use a “linear” description of formal liftings over W of a K3 surface over k. Let Y/W be a proper flat formal scheme such that $Y_k = Y \otimes k$ is a K3 surface. Then $H^2_{\mathrm{DR}}(Y/W)$ is canonically identified with $H = H^2(Y_k/W)$, and $\mathrm{Fil}^2 H^2_{\mathrm{DR}}(Y/W)$ is a line (by definition, a direct summand of rank 1) in H, isotropic (i.e. contained in its orthogonal), lifting $\mathrm{Fil}^2 H^2_{\mathrm{DR}}(Y_k/k)$.

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THEOREM 2.1.11. Suppose $p \neq 2$. Let Y_0 be a K3 surface over k. Set $H = H^2(Y_0/W)$. The map assigning to a proper flat formal scheme Y/W lifting Y_0 the line $\mathrm{Fil}^2 H^2_{\mathrm{DR}}(Y/W)$ is a bijection from the set of liftings of Y_0 over W (i.e. the W-valued points of the formal moduli scheme of Y_0) to the set of liftings of $\mathrm{Fil}^2 H^2_{\mathrm{DR}}(Y_0/k)$ as an isotropic line in H.

First observe that the two types of liftings under consideration (liftings of Y_0 and liftings of $\mathrm{Fil}^2 H^2_{\mathrm{DR}}(Y_0/k)$) are rigid, i.e. have no infinitesimal automorphisms. Let Y_n be a lifting of Y_0 over W_{n+1}. The set of liftings of Y_n over W_{n+2} is a torsor under $H^1(Y_0, T_{Y_0/k})$. On the other hand, let L_n be an isotropic line in $H \otimes W_{n+1}$ lifting $L_0 = \mathrm{Fil}^2 H^2_{\mathrm{DR}}(Y_0/k)$. Then the set of liftings of L_n as an isotropic line in $H \otimes W_{n+2}$ is a torsor under $\mathrm{Hom}(L_0, L_0 \perp / L_0)$: without the isotropy condition one would obtain a torsor under $\mathrm{Hom}(L_0, H_0/L_0)$, where $H_0 = H \otimes k = H^2_{\mathrm{DR}}(Y_0/k)$; since $p \neq 2$, imposing isotropy on the lifting gives the stated result (if $H \otimes W_{n+2} = L_{n+1} \oplus M_{n+1}$, with L_{n+1} isotropic and lifting L_n, an isotropic line lifting L_n is generated by a vector $x + p^{n+1} y$, where x is a basis of L_n, and one must have $2\langle x, y\rangle = 0 \bmod p$). Now $L_0 \perp\ = \mathrm{Fil}^1 H^2(Y_0/k)$, hence

$$\mathrm{Hom}(L_0, L_0 \perp / L_0) = \mathrm{Hom}(\mathrm{Fil}^2 H^2_{\mathrm{DR}}(Y_0/k), \mathrm{gr}^1 H^2_{\mathrm{DR}}(Y_0/k)),$$

and, by (IV 2.4), there is a canonical isomorphism (given by the cup product)

$$H^1(Y_0, T_{Y_0/k}) \simeq \mathrm{Hom}(\mathrm{Fil}^2 H^2_{\mathrm{DR}}(Y_0/k), \mathrm{gr}^1 H^2_{\mathrm{DR}}(Y_0/k)).$$

The conclusion of 2.1.11 follows readily.

REMARK 2.1.12. Note the analogy between 2.1.11 and the theory of liftings of p-divisible groups and abelian varieties (lifting the Hodge filtration in the “crystal” extension

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universal) [14].

Proof of 2.1.9. In view of 2.1.11, it is enough to observe that $e_q^* X$, and hence e_q, is completely characterized by the second equality in (2.1.8.5).

We shall now see that 2.1.7 gives a canonical formal-group structure over W on the formal moduli space S = spf(A), with origin the canonical lifting X_W (2.1.9). For this we shall need the following lemma:

LEMMA 2.1.13. Let $(a', b'_1, \dots, b'_{20}, c')$ be a basis of H and let $(q'_i)_{1 \le i \le 20}$ be a family of elements of A satisfying the same conditions (i), (ii), and (iii) of 2.1.7 as $(a, (b_i), c)$ and (q_i). Then there exist $\alpha \in \mathbb{Z}_p^*$ and $\beta = (\beta_{\mathrm{ij}}) \in \mathrm{GL}(20, \mathbb{Z}_p)$ such that

$$(2.1.13.1) \quad \{a' = \alpha a, \quad c' = c/\alpha,$$
$$q'_i = \prod_{1 \le j \le 20} q_j^{(\beta_{\mathrm{ji}}/\alpha)},$$
$$b'_i = \sum_{1 \le j \le 20} \beta_{\mathrm{ji}} b_j.\}$$

First, since $(a', (b'_i), c')$ is a basis of H adapted to the decomposition $H = H_0 \oplus H_1 \oplus H_2$ and $\langle a', c' \rangle = 1$, there exist $\alpha \in A^*$ and $\beta = (\beta_{\mathrm{ij}}) \in$ GL(20,A) such that

$$a' = \alpha a, \quad c' = c/\alpha, \quad b'_i = \sum_{1 \le j \le 20} \beta_{\mathrm{ji}} b_j.$$

By 2.1.9, write $e = e_q = e_{q'} : A \to W$ for the lifting (2.1.8.3). Since $\nabla a = \nabla a' = 0$, one has $d\alpha = 0$, so α is “constant,” with value $e^*\alpha \in W$. Let φ' be the lifting of Frobenius such that $\varphi'(q'_i) = {q'_i}^p$. One has

$$a' = F(\varphi')\varphi'^* a' = F(\varphi)\varphi^* a = \alpha^\sigma F(\varphi)\varphi^* a = \alpha^\sigma a = \alpha a,$$

hence $\alpha^\sigma = \alpha$, i.e. $\alpha \in \mathbb{Z}_p$. On the other hand,

$$\nabla b'_i = \sum_j (d\beta_{\mathrm{ji}}) \otimes b_j + \sum_j \beta_{\mathrm{ji}} (d \log q_j) \otimes a$$
$$= (d \log q'_i) \otimes a' = (d \log q'_i) \otimes \alpha a,$$

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hence

$$(*) \qquad d\beta_{\mathrm{ji}} = 0 \qquad \text{for every } i, j,$$

and

$$(**) \quad \alpha\, d \log q'_i = \textstyle\sum_j \beta_{\mathrm{ji}}\, d \log q_j \qquad (1 \le i \le 20).$$

Formula (*) means that β_{ji} is constant, with value $e^*\beta_{\mathrm{ji}} \in W$. But
$p\, b'_i = F(\varphi')\varphi'^* b'_i = \sum_j \beta_{\mathrm{ji}}^{\sigma}\, F(\varphi')\varphi'^* b_j$.
Applying e^* and taking account of

$$e^* F(\varphi')\varphi'^* b_j = e^* F(\varphi)\varphi^* b_j = e^* p\, b_j,$$

one obtains $\beta_{\mathrm{ji}}^{\sigma} = \beta_{\mathrm{ji}}$ for every i, j, and hence $\beta \in \mathrm{GL}(20, \mathbb{Z}_p)$. From (**) one deduces that there is a $C \in W^*$ such that

$$q'_i = C \textstyle\prod_j q_j^{(\beta_{\mathrm{ji}}/\alpha)}.$$

Since $e^* q'_i = e^* q_i = 1$, one has $C = 1$, which completes the proof.

THEOREM 2.1.14. Suppose $p \neq 2$. Let X_k be an ordinary K3 surface over k, and let $S = \mathrm{Spf}(A)$ be the corresponding formal moduli space over W. Let G be the formal torus over $\mathbb{Z}_p$ whose character group is $\mathrm{Hom}_{\mathbb{Z}_p}(P_0, P_1)$, with the notation of 2.1.8. There is a canonical isomorphism between S and G_W under which the canonical lifting $X_W \in S$ (2.1.9) corresponds to the identity element $1 \in G_W$.

Let a = $(a, b, (q_i))$ be as in 2.1.7. The family (q_i) gives an isomorphism $u_a : S \simeq (\widehat{G}_m)^{20}_W = \mathrm{Spf}(W[[T_i - 1]])$, $T_i \mapsto q_i$, while the basis (a, b) gives an isomorphism $v_a : \mathbb{Z}_p^{20} \simeq \mathrm{Hom}(P_0, P_1)$, such that $v_a(x_1, \dots, x_{20})(a) = \sum x_i b_i$. If a′ = $(a', b', (q'_i))$ is another choice, there exist $\alpha \in \mathbb{Z}_p^*$ and $\beta = (\beta_{\mathrm{ij}}) \in \mathrm{GL}(20, \mathbb{Z}_p)$ satisfying (2.1.13.1). Write

$$f : (\widehat{G}_m)^{20}_{\mathbb{Z}_p} \simeq (\widehat{G}_m)^{20}_{\mathbb{Z}_p}$$

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for the isomorphism given by $T_i \mapsto \prod_j T_j^{(\beta_{\mathrm{ji}}/\alpha)}$. The isomorphism

$$\mathrm{Hom}(f, \widehat{G}_m) : \mathbb{Z}_p^{20} \simeq \mathbb{Z}_p^{20}$$

induced by f on character groups is multiplication by the matrix β/α. The formulas (2.1.13.1) give commutative squares

$$\begin{array}{ccccc}
u_a & v_a & & & \\
S & \simeq \to (\widehat{G}_m)^{20} & \mathbb{Z}_p^{20} & \simeq \to & \mathrm{Hom}(P_0, P_1) \\
\| & \downarrow f_W & \uparrow \beta/\alpha & & \| \\
S & \simeq \to (\widehat{G}_m)^{20} & \mathbb{Z}_p^{20} & \simeq \to & \mathrm{Hom}(P_0, P_1). \\
u_{a'} & v_{a'} & & &
\end{array}$$

In other words, the isomorphism $S \simeq G_W$ given by (u_a, v_a) is independent of a; it is the asserted canonical isomorphism. By definition X_W corresponds to the augmentation $q_i \mapsto 1$, and 2.1.14 is proved.

DEFINITION 2.1.15. A system a $= (a, b, c, q)$ satisfying the conditions of 2.1.7 will be called a system of canonical coordinates (or parameters) on S.

Observe that the choice of a corresponds to the choice of a basis of P_0 (and is therefore defined up to a factor in $\mathbb{Z}_p^*$) and determines c; on the other hand, once a has been chosen, the datum of b corresponds to the choice of a basis of P_1 (and is therefore defined up to a factor in $\mathrm{GL}(20, \mathbb{Z}_p)$) and determines the isomorphism $(q_i) : S \simeq (\widehat{G}_m)_W^{20}$.

Problems 2.1.16. a) Can one give an "intrinsic" definition of the formal-group structure on S constructed in 2.1.14, for example by exhibiting it as the representative of a suitable functor? (*)

b) Extend the theory of canonical coordinates to the case $p = 2$. This presents several difficulties. First, of course, one must find an adequate substitute for the definition of the q_{ij} in 1.4.2. On the other hand, in the K3 case there are additional difficulties connected with the fact that it is not known, for $p = 2$, whether the quadra-

(*) (added in January 1981), this question has just been answered affirmatively by Nygaard.

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tic form $\langle x, x\rangle$ on $H^2_{\mathrm{DR}}(X_k/k)$ is identically zero (recall that, if X is a K3 surface over the complex numbers, the form $\langle x, x\rangle$ on $H^2(X, \mathbb{Z})$ is even [17]). It can nevertheless be shown [4] that, for Y_0/k ordinary, or more generally if the kernel of the form $\langle x, x\rangle$ on $H^2_{\mathrm{DR}}(Y_0/k)$ is not equal to $\mathrm{Fil}^1 H^2_{\mathrm{DR}}(Y_0/k)$, the map considered in 2.1.11, assigning to a formal lifting Y the isotropic line $\mathrm{Fil}^2 H^2_{\mathrm{DR}}(Y/W) \subset H^2(Y_0/W)$, is surjective with finite fibers: for Y_0 ordinary, the line $W \otimes P_2$ of (2.1.8.5) then defines a finite family of “canonical liftings” of Y_0.

2.2. Liftings of invertible sheaves.

2.2.1. Let $X/S = \mathrm{Spf}(A)$ be the universal formal deformation of a K3 surface X_k/k, and let L_0 be a nontrivial invertible sheaf on X_k. It is known (IV 1.6) that there is a largest closed formal subscheme $T = \Sigma(L_0) \subset S$ such that L_0 extends to an invertible sheaf L on X_T (*), and that T is defined by an equation $f = 0$ with p not dividing f. Assume $p \neq 2$ and X_k ordinary. We shall show that f can then be made explicit in terms of the crystalline Chern class

$$(2.2.1.1) \qquad c_1(L_0) \in H^2(X_k/k).$$

Recall (IV 2.9) that, with the notation of (2.1.8.1),

$$(2.2.1.2) \qquad c_1(L_0) \in P_1.$$

In particular, if (a, b, c, q) is a system of canonical coordinates on S (2.1.15), and $e : A \to W$ is the augmentation given by $q_i \mapsto 1$, one may write

$$(2.2.1.3) \qquad c_1(L_0) = \sum_{1\le i\le 20} x_i(e^* b_i),$$

with $x_i \in \mathbb{Z}_p$. We then have the following result, whose proof will occupy the remainder of the exposé:

(∗) $X_{S'}$ denotes the formal scheme obtained from X by the base change $S' \to S$.

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THEOREM 2.2.2. With the hypotheses and notation of 2.2.1, T is defined by the equation

$$(2.2.2.1) \qquad \prod_{1\leq i\leq 20} q_i^{x_i} = 1.$$

In other words, if, by means of the basis e^*a of P_0, $c_1(L_0)$ is identified with a character x of the formal torus S (2.1.14), T is the formal subgroup defined by

$$(2.2.2.2) \qquad T = \mathrm{Ker}(x).$$

In particular, if p does not divide $c_1(L_0)$ (which also means [16, 1.4] that p does not divide the class of L_0 in $\mathrm{NS}(X_k)$), T is a smooth formal subtorus over W of relative dimension 19.

2.2.3. Let us first indicate how one may heuristically convince oneself of the validity of 2.2.2. Over the complex numbers, T would be the locus where the horizontal section of H^2_{DR} through $c_1(L_0)$ remains of type (1,1), i.e. lies in $\mathrm{Fil}^1\, H^2_{\mathrm{DR}}$. Let $\underline{x} \in H^2_{\mathrm{DR}}(X/S) \otimes K[[q_i - 1]]$ be the horizontal section such that $e^*\underline{x} = c_1(L_0)$. An immediate calculation from (2.1.7 (ii)) gives

$$(2.2.3.1) \qquad \underline{x} = \left(-\sum_{1\leq i\leq 20} x_i \log q_i\right) a + \sum_{1\leq i\leq 20} x_i b_i.$$

Thus, heuristically, $-\sum x_i \log q_i = 0$, "i.e." $\prod q_i^{x_i} = 1$, is the desired equation. Naturally this argument is insufficient, but it can be adapted by using the crystalline Chern class to control, step by step, the obstruction to extending L_0.

For this purpose we shall need some reminders concerning Chern classes and obstructions, supplementing (IV 2.8, 2.9).

2.2.4. Let $i : Y_0 \hookrightarrow Y$ be a closed immersion of schemes (or formal schemes) defined by an ideal I, and let E_0 be an invertible sheaf on Y_0, of class $c\ell(E_0) \in H^1(Y_0, \mathcal{O}^*_{Y_0})$. The exact sequence of abelian sheaves on Y

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$$(2.2.4.1) \qquad 0 \to (1+I)^* \to \mathcal{O}_Y^* \to \mathcal{O}_{Y_0}^* \to 0$$

shows that the obstruction $\omega^*(E_0, i)$ to extending E_0 to an invertible sheaf on Y is the image of $c\ell(E_0)$ under the connecting morphism in the exact cohomology sequence deduced from (2.2.4.1):

$$(2.2.4.2) \qquad \omega^*(E_0, i) = d\, c\ell(E_0) \in H^2(Y, (1+I)^*).$$

If $Y \to Z$ is a morphism, one may consider the “multiplicative” de Rham complex

$$\Omega^{\bullet *}_{Y/Z} = (\mathcal{O}_Y^* \xrightarrow{\mathrm{dlog}} \Omega^1_{Y/Z} \xrightarrow{d} \Omega^2_{Y/Z} \xrightarrow{d} \cdots),$$

and (2.2.4.1) refines to an exact sequence of complexes

$$(2.2.4.3) \qquad 0 \to (1+I)\Omega^{\bullet *}_{Y/Z} \to \Omega^{\bullet *}_{Y/Z} \to \mathcal{O}_{Y_0}^* \to 0,$$

where

$$(1+I)\Omega^{\bullet *}_{Y/Z} \quad := \quad ((1+I)^* \xrightarrow{\mathrm{dlog}} \Omega^1_{Y/Z} \xrightarrow{d} \Omega^2_{Y/Z} \xrightarrow{d} \cdots).$$

The obstruction $\omega^*(E_0, i)$ is the image, under the natural map, of the class

$$(2.2.4.4) \qquad c^*(E_0, i, Y/Z) \in H^2(Y, (1+I)\Omega^{\bullet *}_{Y/Z})$$

obtained from $c\ell(E_0)$ as the connecting image in the exact cohomology sequence associated with (2.2.4.3).

2.2.5. Assume $I^2 = 0$. Then

$$(1+I)^* = 1+I \quad \simeq \quad I, \qquad \log(1+x) = x,$$

and similarly $(1+I)\Omega^{\bullet *}_{Y/Z}$ is identified with

$$I\Omega^{\bullet}_{Y/Z} \quad := \quad (I \xrightarrow{d} \Omega^1_{Y/Z} \xrightarrow{d} \Omega^2_{Y/Z} \xrightarrow{d} \cdots).$$

Write

$$(2.2.5.1) \qquad \omega(E_0, i) \in H^2(Y, I), \qquad c(E_0, i, Y/Z) \in H^2(Y, I\Omega^{\bullet}_{Y/Z})$$

for the images of $\omega^*(E_0, i)$ and $c^*(E_0, i, Y/Z)$ under these identifications. The obstruction $\omega(E_0, i)$ is again the image of $c(E_0, i, Y/Z)$ under the natural map.

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Assume further that $Z = (Z, K, \delta)$ is a PD-scheme on which p is nilpotent [1], or that Z is a P-adic base (with $p \in P$) in the sense of [3, 7.17]: the case actually of interest to us is that in which Z is a p-adically complete formal W-scheme, endowed with the standard divided powers on $p\mathcal{O}_Z$. If Y is smooth over Z, and if the trivial divided powers γ on I (i.e. $\gamma^i(x) = 0$ for i>1) are compatible with those of Z, then

$$H^2(Y, I\Omega^{\bullet}_{Y/Z}) = H^2(Y_0/Z, J_{Y_0/Z})$$

(where $J_{Y_0/Z}$, as usual, denotes the crystalline ideal that is the kernel of $\mathcal{O}_{Y_0/Z} \to \mathcal{O}_{Y_0}$), and the class $c(E_0, i, Y/Z)$ of (2.2.5.1) is precisely the crystalline Chern class of L_0 relative to Y_0/Z, defined in [2]:

$$(2.2.5.2) \qquad c(E_0, i, Y/Z) = c_1(E_0)_{Y_0/Z} \in H^2(Y_0/Z, J_{Y_0/Z}).$$

We shall apply these remarks when there are Cartesian squares

$$\begin{tikzcd} Y_0 \arrow[r, "i"] \arrow[d, "f_0"] & Y \arrow[r] \arrow[d, "f"] & X \arrow[d] \\ Z_0 \arrow[r, "j"] & Z \arrow[r] & S, \end{tikzcd}$$

(2.2.5.3)

where X/S is the universal formal deformation of X_k, Z is an affine p-adically complete formal W-scheme, and j is a closed immersion defined by a square-zero ideal J, so that $I = f^*(J)$. Since $R^1 f_{0*}(\mathcal{O}_{Y_0}) = 0$ and hence $H^1(Y_0, \mathcal{O}) = 0$, the canonical maps

$$H^2(Y, I) \to H^2(Y, \mathcal{O}), \qquad H^2(Y, I\Omega^{\bullet}_{Y/Z}) \to H^2_{\mathrm{DR}}(Y/Z)$$

are injective; we shall therefore regard $\omega(E_0, i)$ (respectively $c(E_0, i, Y/Z)$) as an element of $H^2(Y, \mathcal{O})$ (respectively $H^2_{\mathrm{DR}}(Y/Z)$). Moreover, if $p \neq 2$, or if $p\mathcal{O}_Z = 0$, the trivial divided powers on I are compatible with the standard divided powers on $p\mathcal{O}_Z$. From what has

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just been seen, the obstruction $\omega(E_0, i)$ is therefore given by the following recipe:

LEMMA 2.2.6. In the situation of (2.2.5.3), if $p \neq 2$, or if $p\mathcal{O}_Z = 0$, the obstruction $\omega(E_0, i)$ to extending E_0 to an invertible sheaf on Y is the image, under the canonical map $H^2_{\mathrm{DR}}(Y/Z) \to H^2(Y, \mathcal{O}_Y)$, of the crystalline Chern class of E_0 relative to Z, $c_1(E_0)_{Y_0/Z} \in H^2_{\mathrm{DR}}(Y/Z)$.

In other words, under the hypotheses of 2.2.6, E_0 extends to an invertible sheaf on Y if and only if

$$c_1(E_0)_{Y_0/Z} \in \mathrm{Fil}^1 H^2_{\mathrm{DR}}(Y/Z).$$

2.2.7. The usefulness of the recipe in 2.2.6 is that it gives a computational principle for the Chern class $c_1(E_0)_{Y_0/Z}$, which we now recall.

Under the hypotheses of 2.2.6, one may consider the Chern class

$$(2.2.7.1) \qquad c_1(E_0)_{f_0/W} \in H^2(Y_0/W),$$

and its image

$$(2.2.7.2) \qquad c_1(E_0)_{f_0/W} \in H^0(Z_0/W, R^2 f_{0*}(\mathcal{O}_{Y_0/W}))$$

under the canonical map

$$H^2(Y_0/W) \to H^0(Z_0/W, R^2 f_{0*}(\mathcal{O}_{Y_0/W})).$$

On the other hand, for every PD-thickening Z_1 of Z_0 one may consider the Chern class

$$(2.2.7.3) \qquad c_1(E_0)_{Y_0/Z_1} \in H^2(Y_0/Z_1).$$

The relation between (2.2.7.2) and (2.2.7.3) is as follows: the crystal $R^2 f_{0*}(\mathcal{O}_{Y_0/W})$ has a “value” $R^2 f_{0*}(\mathcal{O}_{Y_0/W})(Z_1)$ on Z_1, and, for affine Z_1, the section

$$c_1(E_0)_{f_0/W}(Z_1) \in H^0(Z_1, R^2 f_{0*}(\mathcal{O}_{Y_0/W})(Z_1)) = H^2(Y_0/Z_1)$$

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defined by (2.2.7.2) is (2.2.7.3). Suppose now that the map $Z_0 \to S$ in (2.2.5.3) factors as

$$(2.2.7.4) \quad Z_0 \xrightarrow{j'} Z' \to S,$$

where Z' is a smooth, p-adically complete formal W-scheme and j' is a closed immersion. Let $\widehat{Z'} = D_{Z_0}(Z')$ be the p-adically completed PD-envelope of Z_0 in Z'. By Berthelot [1] (or [3, §7]), the crystal $R^2 f_{0*}(\mathcal{O}_{Y_0/W})$ is described by an $\mathcal{O}_{\widehat{Z'}}$ − module with integrable connection relative to W. In the present case, since there are Cartesian squares

$$\begin{array}{ccccc} Y_0 & \longrightarrow & \widehat{Y'} & \longrightarrow & X \\ \downarrow & & \downarrow & & \downarrow \\ Z_0 & \longrightarrow & \widehat{Z'} & \longrightarrow & S, \end{array}$$

(2.2.7.5)

this module is none other than $H^2_{\mathrm{DR}}(X/S) \otimes \mathcal{O}_{\widehat{Z'}}$, equipped with the Gauss-Manin connection. The class (2.2.7.2) is interpreted as a horizontal section of this module. When, in the situation of (2.2.5.3), Z is a closed subscheme of Z', the universal property of PD-envelopes maps Z into $\widehat{Z'}$, and the section of $H^2_{\mathrm{DR}}(X/S) \otimes \mathcal{O}_Z = H^2_{\mathrm{DR}}(Y/Z)$ induced by (2.2.7.2) is precisely $c_1(E_0)_{Y_0/Z}$. We shall see below how to use the horizontality of the section (2.2.7.2) of $H^2_{\mathrm{DR}}(X/S) \otimes \mathcal{O}_{\widehat{Z'}}$ to calculate it when E_0 extends the given sheaf L_0 on X_k, using suitably chosen embeddings j' as in (2.2.7.4).

2.2.8. For this purpose we shall need one last, somewhat technical, ingredient concerning PD-envelopes. Write

$$(2.2.8.1) \qquad W\langle\langle t\rangle\rangle = W\langle\langle t_1, \ldots, t_n\rangle\rangle$$

for the p-adically completed PD-envelope of the ideal $(t_1, \ldots, t_n)$ of W[[t]] = $W[[t_1, \ldots, t_n]]$: it is the subring of K[[t]] consisting of series $\sum a_i t^i / i!$ with $a_i \in W$ tending to 0. Now consider

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the ideal $t^r = (t_1^{r_1}, \dots, t_n^{r_n})$ of W[[t]], and the p-adically completed PD-envelope of t^r in W[[t]]:

$$(2.2.8.2) \qquad D_{(t^r)}(W[[t]]).$$

Using the compatibility of the formation of PD-envelopes with flat extensions [1, I 2.7.4] for the finite flat morphism W[[t]] → W[[t]], $t_i \mapsto t_i^{r_i}$, one obtains the following description of (2.2.8.2): its underlying additive group consists of formal series, with coefficients in W tending to 0, in the $t^{(m)}$, where

$$t^{(m)} = t^s (t^r)^{[q]}, \qquad \text{if } m = rq + s, \qquad 0 \le s < r,$$

and multiplication is the evident multiplication formally given by $(t^r)^{[q]} = t^{rq}/q!$.
The map

$$(2.2.8.3) \qquad D_{(t^r)}(W[[t]]) \to W\langle\langle t \rangle\rangle$$

defined by the inclusion $(t^r) \subset (t)$ sends the basis element $t^{(m)}$ of multi-degree m to $(m!/q!)t^{[m]}$; in particular, it is injective.
2.2.9. Proof of 2.2.2. Since L_0 is nontrivial, it is known (IV 3.4) that $c_1(L_0) \neq 0$. Let p^m be the greatest power of p dividing $x = c_1(L_0)$, so that $x = p^m y$ with $p \nmid y$. Thus y is part of a basis of the character group of S, and the coordinates (a, b, q) may be chosen so that $y = (1, 0, \dots, 0)$, i.e. $x = p^m(e^* b_1)$. Put

$$T' = \mathrm{Spf}(W[[q_i - 1]]/(q_1^{p^m} - 1)).$$

We must prove that $T = T'$. We proceed in four steps.
a) Extension of L_0 over $X_{\mathrm{Spec}(k[q_1-1]/(q_1^{p^m}-1))}$. This can be done in two ways. The quicker is to observe that, by [16, 1.4], since p^m divides $c_1(L_0)$, p^m also divides the class of L_0 in $\mathrm{NS}(X_k)$. If F denotes Frobenius on X_k, L_0 is therefore the inverse image under F^m of an invertible sheaf L_0', and

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consequently L_0 extends to every infinitesimal k-neighborhood of X_k of order $\leq\ p^m - 1$. One may also proceed directly, extending L_0 step by step over X_{Y_n}, where $Y_n = \mathrm{Spec}(k[t_1]/t_1^n)$ and $q_1 - 1 = t_1$. Assume $m > 0$ and that L_0 has been extended to an invertible sheaf L_{Y_n} on X_{Y_n} for $n < p^m$; we show that L_{Y_n} extends to an invertible sheaf on $X_{Y_{n+1}}$. Let $c(L_{Y_n})_{Y_{n+1}} \in\ H^2_{\mathrm{DR}}(X_{Y_{n+1}}/Y_{n+1})$ be the crystalline Chern class of L_{Y_n}. By 2.2.6, it is enough to show that the image of this class in $H^2(X_{Y_{n+1}}, \mathcal{O})$ is zero. Now Y_n is the formal subscheme of $\mathrm{Spf}(W[[t_1]])$ defined by the ideal (p, t_1^n), and, by 2.2.7, if D_n denotes the p-adically completed PD-envelope of this ideal in $W[[t_1]]$, i.e. $D_n = D_{(t_1^n)}(W[[t_1]])$, then $c(L_{Y_n})_{Y_{n+1}}$ is induced by the horizontal section $c(L_{Y_n})_{D_n} \in H^2_{\mathrm{DR}}(X/S) \otimes D_n$ of the type in (2.2.7.2). Since, by 2.2.8, the map $D_n \to\ D_1 = W\langle\langle t_1\rangle\rangle$ is injective, $c(L_{Y_n})_{D_n}$ is determined by its image in $H^2_{\mathrm{DR}}(X/S) \otimes D_1$. That image is precisely the horizontal section $c(L_0)_{D_1}$ defined by L_0: this follows from the functoriality of the class (2.2.7.1), in view of the commutative diagram

$$\begin{array}{ccccccc}
Y_n & \longrightarrow & \mathrm{Spf}(W[[t_1]]) & & & & \\
 & & \downarrow & & \uparrow\ \nearrow & & \downarrow \\
 & & \downarrow & & \downarrow & & \\
\mathrm{Spec}(k) & \longrightarrow & \mathrm{Spec}(W) & & & &
\end{array}$$

and the fact that L_{Y_n} extends L_0. Since $c(L_0)$ is the horizontal class extending $c_1(L_0) = p^m e^* b_1$, an immediate calculation using 2.1.7 gives

$$c(L_0)_{D_1} = -(p^m \log q_1) a_{D_1} + p^m (b_1)_{D_1}$$

(where $(a_{S'}, b_{S'})$ denotes the inverse image of (a, b) on S' for $S' \to\ S$). Hence, by the injectivity of $D_n \to\ D_1$,

$$c(L_{Y_n})_{D_n} = -(\log q_1^{p^m}) a_{D_n} + p^m (b_1)_{D_n}$$

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(observe that $q_1^{p^m} - 1$ belongs to the ideal generated by p and $t_1^n = (q_1 - 1)^n$, so that the logarithm is meaningful by virtue of the divided powers on (p, t_1^n)). The map $D_n \to \mathcal{O}_{Y_{n+1}}$ defined by the universal property of PD-envelopes sends $x^{[i]}$ (for $x \in (p, t_1^n)$) to x mod (p, t_1^{n+1}) for $i = 1$ and to 0 for $i \geq 2$. In particular, it sends $\log q_1^{p^m}$ to $(q_1^{p^m} - 1)$ mod $(p, t_1^{n+1}) = 0$, since $n < p^m$. In other words, the image $c(L_{Y_n})_{Y_{n+1}} \in H^2_{\mathrm{DR}}(X/S) \otimes \mathcal{O}_{Y_{n+1}}$ of $c(L_{Y_n})_{D_n}$ lies in Fil^1; therefore, by what was said above, L_{Y_n} extends to an invertible sheaf on $X_{Y_{n+1}}$.

b) Extension of L_0 over X_Z, where $Z = \mathrm{Spf}(W[[q_1 - 1]]/(q_1^{p^m} - 1))$. Let L_Y denote the extension, obtained in a), of L_0 over X_Y. Put

$$Z_n = \mathrm{Spec}(W_n[[q_1 - 1]]/(q_1^{p^m} - 1)).$$

Assume that L_Y has been extended to an invertible sheaf L_{Z_n} on X_{Z_n}; we show that L_{Z_n} extends to an invertible sheaf on $X_{Z_{n+1}}$. By 2.2.6 (which applies because $p \neq 2$), it is enough to verify that the Chern class $c(L_{Z_n})_{Z_{n+1}} \in H^2_{\mathrm{DR}}(X_{Z_{n+1}}/Z_{n+1})$ lies in Fil^1. It is induced by the class $c(L_{Z_n})_D \in H^2_{\mathrm{DR}}(X_D/D)$, where D is the p-adically completed PD-envelope of Y_n in $W[[q_1 - 1]]$. If $D_I(\ )$ denotes the p-adically completed PD-envelope of the ideal I, then

$$\begin{aligned} D &:= D_{(p^n, q_1^{p^m} - 1)}(W[[q_1 - 1]]) \\ &= D_{(p, q_1^{p^m} - 1)}(W[[q_1 - 1]]) \\ &= D_{(p, (q_1 - 1)^{p^m})}(W[[q_1 - 1]]) \\ &= D_{(t_1^{p^m})}(W[[t_1]]) \end{aligned}$$

where $t_1 = q_1 - 1$. By 2.2.8, the map $D \to D_{(t_1)}(W[[t_1]]) = W\langle\langle t_1 \rangle\rangle$ defined by the inclusion $(p^n, q_1^{p^m} - 1) \subset (p, q_1 - 1)$ is therefore injective, and the same argument as in a) shows that

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$$c(L_{Z_n})_D = -(\log q_1^{p^m})a_D + p^m(b_1)_D.$$

Consequently, $c(L_{Z_n})_{Z_{n+1}}$, the image of $c(L_{Z_n})_D$ under the map $D \to \mathcal{O}_{Z_{n+1}}$ defined by the universal property of PD-envelopes (which sends $x^{[i]}$ (for $x \in (p^n, q_1^{p^m} - 1)$) to the image of x in $\mathcal{O}_{Z_{n+1}}$ for $i = 1$ and to 0 for $i > 1$), is

$$\begin{aligned} c(L_{Z_n})_{Z_{n+1}} &= -(q_1^{p^m} - 1)a_{Z_{n+1}} + p^m(b_1)_{Z_{n+1}} \\ &= p^m(b_1)_{Z_{n+1}} \end{aligned}$$

(since $q_1^{p^m} - 1 = 0$ on Z_{n+1}). Thus $c(L_{Z_n})_{Z_{n+1}}$ belongs to $\mathrm{Fil}^1\, H^2_{\mathrm{DR}}(X_{Z_{n+1}}/Z_{n+1})$, and consequently L_{Z_n} extends to an invertible sheaf on $X_{Z_{n+1}}$. Hence L_Y extends to an invertible sheaf L_Z on X_Z.

c) Extension of L_0 over $X_{T'}$. Put $t_i = q_i - 1$. Let $n = (n_2, \ldots, n_{20})$ be a sequence of integers ≥ 1, and put

$$T'_n = \mathrm{Spf}(W[[t]]/(q_1^{p^m} - 1, t_2^{n_2}, \ldots, t_{20}^{n_{20}}))$$

(so $T'_{(1,\ldots,1)} = Z$). Assume L_Z has been extended to an invertible sheaf $L_{T'_n}$ on $X_{T'_n}$. By an argument analogous to that used in b), if $2 \leq i \leq 20$ and $n + 1_i = (n_2, \ldots, n_{i-1}, n_i + 1, n_{i+1}, \ldots, n_{20})$, then $L_{T'_n}$ extends to an invertible sheaf on $X_{T'_{(n+1_i)}}$. Induction therefore shows that L_Z extends to an invertible sheaf $L_{T'}$ on $X_{T'}$.

d) End of the proof. To prove $T' = T$, it remains to check that if $T'' \supset T'$ is a closed formal subscheme of S, defined by an ideal I, such that $L_{T'}$ extends to an invertible sheaf on $X_{T''}$, then $T'' = T'$. Equivalently, if $T'' \neq T'$ and $I^2 = 0$, the obstruction to extending $L_{T'}$ to an invertible sheaf on $X_{T''}$ must be nonzero, i.e. the Chern class $c(L_{T'})_{T''} \in H^2_{\mathrm{DR}}(X_{T''}/T'')$ must not belong to Fil^1. But the same

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calculation as before gives

$$c(L_{T'})_{T''} = -(q_1^{p^m} - 1)a_{T''} + p^m(b_1)_{T''}.$$

Since $T'' \neq T'$, the image of $q_1^{p^m} - 1$ in T'' is nonzero; hence $c(L_{T'})_{T''} \notin \mathrm{Fil}^1$, which completes the proof.

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APPENDIX TO EXPOSE V

Nicholas M. Katz

A1. UNIQUENESS OF GROUP STRUCTURES

Let k be a perfect field, $W = W(k)$ its ring of Witt vectors, and $\sigma : W \cong \to W$ the absolute Frobenius automorphism of W. Let M be a finite-dimensional formal Lie variety over W, i.e. $M = \mathrm{Spf}(A)$ with A non-canonically isomorphic to $W[[T_1, \dots, T_n]]$, $n = \dim M$. Suppose we are given a W-morphism of formal Lie varieties

$$\Phi : M \to M^{(\sigma)}$$

whose reduction modulo p is the absolute Frobenius endomorphism

$$\mathrm{Frob} : M \otimes_W k \to (M \otimes_W k)^{(\sigma)}.$$

UNIQUENESS LEMMA A1.1.(1) Given (M, Φ) as above, there exists at most one structure of commutative formal Lie group over W on the formal Lie variety M for which the given map $\Phi : M \to M^{(\sigma)}$ is a group homomorphism. (2) If this structure exists, it makes M into a toroidal formal group, and the given $\Phi : M \to M^{(\sigma)}$ is the unique group homomorphism lifting Frobenius. (3) If (M_1, Φ_1) and (M_2, Φ_2) both admit group structures as in (2) above, then a morphism

$$f : M_1 \to M_2$$

of formal Lie varieties over W is a group homomorphism if and only if the diagram

$$\begin{tikzcd} M_1 \arrow[r, "f"] \arrow[d, "\Phi_1"] & M_2 \arrow[d, "\Phi_2"] \\ M_1^{(\sigma)} \arrow[r, "f^{(\sigma)}"] & M_2^{(\sigma)} \end{tikzcd}$$

commutes.

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PROOF. We begin by proving (2). Thus let G be a finite-dimensional commutative formal Lie group over W, together with a homomorphism

$$\Phi : G \to G^{(\sigma)}$$

which lifts Frobenius. We must show that G is toroidal, and that Φ is unique. By the rigidity of toroidal groups, it suffices to show that $G \otimes k$ is toroidal, and for this it suffices to show that Ker(Frob) is toroidal. For this, we first observe that $\Phi : G \to G^{(\sigma)}$ is finite (because it is finite modulo p, being a lifting of Frobenius) and flat (because it is a finite morphism between regular local rings of the same dimension). Therefore Ker(Φ) is a finite flat commutative group-scheme over W whose reduction mod p is Ker(Frob). According to Fontaine, if we denote by N the contravariant Dieudonné module of Ker(Frob), then the lifting Ker(Φ) is described by a W-submodule $L \subset N$ which satisfies

$$\begin{array}{ll} a) & L/\mathrm{pL} \cong N/\mathrm{FN} \\ b) & V|L \text{ is injective.} \end{array}$$

But N is killed by F, and is of finite length over W. Therefore a) implies that $L = N$, and b) then shows that V is injective, and hence bijective on N. Therefore Ker(Frob) is toroidal, as required.
We next prove (3). By extending scalars $W(k) \to W(\overline{k})$, we may suppose k algebraically closed. Then M_1 and M_2 become isomorphic to products of $\widehat{G}_m$, and our commutative diagram - in the category of formal Lie varieties - becomes

$$\begin{array}{ccc} (\widehat{G}_m)^{n_1} & \xrightarrow{f} & (\widehat{G}_m)^{n_2} \\ \downarrow p & & \downarrow p \\ (\widehat{G}_m)^{n_1} & \xrightarrow{f^{(\sigma)}} & (\widehat{G}_m)^{n_2}. \end{array}$$

If f is a group homomorphism, then $f = f^{(\sigma)}$, and the diagram commutes. To prove the converse, we argue as follows.
In terms of "multiplicative" coordinates T on $(\widehat{G}_m)^{\mathrm{hi}}$, $i = 1,2$, our

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hypothesis on f is:

$$f^{(\sigma)}(T^p) = (f(T))^p, \quad f(1) \equiv 1 \bmod p.$$

Iterating, we find

$$f^{(\sigma^n)}(T^{\mathrm{pn}}) = (f(T))^{\mathrm{pn}},$$

so in particular

$$f^{(\sigma^n)}(1) = (f(1))^{\mathrm{pn}} \to 1 \,\mathrm{as}\, n \to \infty.$$

Therefore f(1) = 1. Now take logarithms, i.e. let

$$F : (\widehat{G}_a)^{n_1} \to (\widehat{G}_a)^{n_2}$$

be the unique pointed morphism (over the fraction field of W) for which

$$F(\log T) = \log(f(T));$$

then we have

$$F^{(\sigma)}(\mathrm{pX}) = \mathrm{pF}(X).$$

Therefore F is linear, and has coefficients in Q_p. As these coefficients are intrinsically the matrix entries of the tangent map of f at the origin, they lie in W as well, hence F is a linear map with coefficients in $Z_p = W \cap Q_p$. Therefore f(T) = exp(F(log T)) is a homomorphism.

Finally, we obtain (1) as the special case $(M_1, \Phi_1) = (M_2, \Phi_2)$, f = id, of (3). Q.E.D.

COROLLARY A1.2. Suppose that k is algebraically closed, and that (M, Φ) admits a group structure as above. Then

(1) The character group $X(M) = \mathrm{Hom}_{W-\mathrm{gp}}(M, \widehat{G}_m)$ is a free $Z_p -$module of rank $n = \dim M$, and the natural map (of functors in $Z_p -$modules)

$$M \to \mathrm{Hom}_{Z_p}(X(M), \widehat{G}_m)$$

is an isomorphism.

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(2) Let q be a function on M (i.e. $q \in A$ where $M = \mathrm{Spf}(A)$) with $q = 1$ modulo the maximal ideal of A. Then $q \in X(M)$ if and only if q transforms under Φ by

$$\Phi^*(q^{(\sigma)}) = q^p.$$

(3) Let ω be a (continuous) one-form on A, i.e. $\omega \in (\Omega^1_{A/W})^{\mathrm{contin}}$. Then $\omega = \mathrm{dq}/q$ with $q \in X(M) = \mathrm{Hom}(M, \widehat{G}_m)$ if and only if

$$\Phi^*(\omega^{(\sigma)}) = p\omega.$$

If ω satisfies this, the associated $q \in X(M)$ is unique; it is given by the formula

$$q(X) = \exp(\textstyle\int_O^X \omega), \quad O = \text{the origin in } M.$$

(4) Let $\tau \in A \widehat{\otimes}_W$ W[1/p] be a function on $M \otimes_W$ W[1/p]. Then $\tau = \log(\mathrm{q})$ for some $q \in X(M)$ if and only if τ satisfies

$$\tau(O) = 0, \quad \Phi^*(\tau^{(\sigma)}) = p\tau.$$

If τ satisfies this, then the associated q is unique; it is given by the formula

$$q = \exp(\tau).$$

(5) Let $q_1, \dots, q_n$ be $n = \dim M$ elements of $X(M)$, $\omega_1, \dots, \omega_n$ the corresponding differentials $\omega_i = \mathrm{dq}_i/q_i$ and $\tau_1, \dots, \tau_n$ the correspon- ding "functions with denominators" $\tau_i = \log q_i$. Then the following conditions are equivalent:

$a)$ $q_1, \dots, q_n$ form a Z_p base of $X(M)$

$b)$ the natural map

$$M \to \mathrm{Spf}(W[[q_1 - 1, \dots, q_n - 1]])$$

is an isomorphism

$c)$ $\omega_1, \dots, \omega_n$ form an $A -$ base of $(\Omega^1_{A/W})^{\mathrm{contin}}$

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$$d)\ \omega_1, \ldots, \omega_n \text{ form } a\ Z_p-\text{base of}$$

$$\{\omega \in \Omega^1 \mid \Phi^*(\omega^{(\sigma)}) = p\omega\}$$

$$e)\ \tau_1, \ldots, \tau_n \text{ form } a\ Z_p-\text{base of}$$

$$\{\tau \in A\widehat{\otimes} W[1/p] \mid \tau(O) = 0,\ \Phi^*(\tau^{(\sigma)}) = p\tau\}.$$

PROOF. Because M is toroidal, and k is algebraically closed, M is (non-canonically) isomorphic to $(\widehat{G}_m)^n$. This makes (1) obvious. Assertion (2) is a particular case of part (3) of the uniqueness lemma, namely $M_1 = M$, $M_2 = \widehat{G}_m$. Assertion (3) becomes obvious if we choose a Z_p-basis $q_1, \ldots, q_n$ of $X(M)$, i.e. if we choose an isomorphism

$$M \cong (\widehat{G}_m)^n,$$

and write ω as an A-linear combination of the differentials dq_i/q_i:

$$\omega = \Sigma f_i \,\mathrm{dq}_i/q_i.$$

The condition

$$\Phi^*(\omega^{(\sigma)}) = p\omega$$

means precisely that the coefficient functions f_i each satisfy

$$\Phi^*(f_i^{(\sigma)}) = f_i,$$

which in turn implies that each coefficient function f_i is simply a constant in Z_p. Therefore

$$\omega = \mathrm{dq}/q \quad \text{for} \quad q = \prod (q_i)^{\mathrm{fi}};$$

because q(O) = 1, we obtain by integration the formula

$$\exp(\textstyle\int_O^X \omega) = q(X).$$

For assertion (4), first note that the Dieudonné-Dwork integrality criterion (cf. 1.4.4) guarantees that the series q, defined as

$$q \quad \text{dfn} = \quad \exp(\tau)$$

is integral (i.e. lies in A) and lies q(O) = 1. From the equality

$$\Phi^*(\tau^{(\sigma)}) = p\tau$$

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(and not simply their congruence modulo pA) we see that

$$\Phi^*(q^{(\sigma)}) = q^p,$$

and we conclude by (2) that q lies in $X(M)$.
In assertion (5), the equivalence of a), b) and c) is physically obvious for $(\widehat{G}_m)^r$, and the equivalence of a) with d) and e) is obvious from parts 3) and 4) above. Q.E.D.

A2. SHARPENINGS OF 1.4

COROLLARY A2.1 OF 1.4.2. With the hypotheses and notations of 1.4.2, suppose that there exists a lifting Φ_{can} of Frobenius on $\mathrm{Spf}(A)$ such that

$$F(\Phi_{\text{can}})(\Phi^*_{\text{can}}(\mathrm{Fil}^1)^{(\sigma)}) \subset \mathrm{Fil}^1.$$

Let O denote the W-valued point of $\mathrm{Spf}(A)$ which is the $\Phi_{\text{can}}-$ Teichmuller representative of the augmentation $A \to k$. Then in formulaire 1.4.2.1-7 we have the further precisions

$$\begin{cases} \nabla a_i = 0 \\ \nabla b_i = \sum_j \eta_{\text{ij}} \otimes a_j \\ F(\Phi_{\text{can}})(\Phi^*_{\text{can}}(a_i^{(\sigma)})) = a_i \\ F(\Phi_{\text{can}})(\Phi^*_{\text{can}}(b_i^{(\sigma)})) = p\, b_i \\ \Phi^*_{\text{can}}(\eta_{\text{ij}}^{(\sigma)}) = p\eta_{\text{ij}}, \quad d\eta_{\text{ij}} = 0 \end{cases}$$

$$\begin{cases} d\tau_{\text{ij}} = \eta_{\text{ij}} \\ \Phi^*_{\text{can}}(\tau_{i,j}^{(\sigma)}) = p\tau_{\text{ij}} \\ \tau_{\text{ij}}(O) = 0 \end{cases}$$

$$\begin{cases} q_{\text{ij}} = \exp(\tau_{\text{ij}}) \text{ is defined, lies in } A, \text{ and} \\ q_{\text{ij}}(O) = 1 \\ \Phi^*_{\text{can}}(q_{\text{ij}}^{(\sigma)}) = q_{\text{ij}}^p. \end{cases}$$

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FURTHER COROLLARY A2.2. (Analogue of (4.7)(1) Given Φ_{can} as above, suppose in addition that (1.4.6.3) : $T_{A/W} \to \mathrm{Hom}_A(\mathrm{Fil}^1, U)$ is an isomorphism. Then $(\mathrm{Spf}(A), \Phi_{\mathrm{can}})$ admits a group structure, and the morphism

$$\mathrm{Spf}(A) \to \prod_{i,j} \widehat{G}_m$$

defined by the $q_{i,j}$ is an isomorphism of groups.

(2) If $p \neq 2$, then Φ_{can} is the unique lifting of Frobenius such that

$$F(\Phi_{\mathrm{can}})(\Phi^*_{\mathrm{can}}(\mathrm{Fil}^1)^{(\sigma)}) \subset \mathrm{Fil}^1 .$$

PROOF. The formulaire is simply obtained from the one given in 1.4.2.1-7 by noting that the $u_{i,j}(\Phi_{\mathrm{can}})$ all vanish. The first assertion of the further corollary has already been proven when $p \neq 2$, but the condition $p \neq 2$ was used only to assure that $\exp(\tau_{i,j}(O))$ make sense. As our hypotheses on Φ_{can} give $\tau_{i,j}(O) = 0$, this problem will not arise, and the proof given goes through tel quel. The unicity of Φ_{can} in case $p \neq 2$ resulte from the observation that the series $q_{\mathrm{ij}} = \exp(\tau_{\mathrm{ij}})$ are then definible without reference to a particular choice of Φ, and furnish an isomorphism

$$M \cong \mathrm{Spf}(W[[q_{\mathrm{ij}} - 1]]);$$

our Φ_{can} is the lifting of Frobenius given by

$$\Phi_{\mathrm{can}}(q_{\mathrm{ij}}^{(\sigma)}) = (q_{\mathrm{ij}})^p . \quad Q.E.D.$$

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A3. FORMAL MODULI OF ORDINARY ABELIAN VARIETIES ; THE SERRE-TATE & DWORK GROUP STRUCTURES

Let k be an algebraically closed field of characteristic $p > 0$, and X_o/k an ordinary abelian variety over k. Let M be its formal deformation space, and X/M the corresponding formal abelian scheme. One knows that M is a g^2-dimensional formal Lie variety over $W = W(k)$. Because X_o/k is ordinary, the formal group $\widehat{X}$ of X/M is non-canonically isomorphic to $(\widehat{G}_m)^g$ over M. The "canonical subgroup" $H_{\text{can}} \subset X$ is defined to be the kernel of $[p]$ in $\widehat{X}$; it is the unique finite flat subgroup-scheme of X/M which mod p becomes the kernel of the relative Frobenius endomorphism of $X \otimes_W k/M \otimes_W k$. We denote by

$$F_{\text{can}} : X \to X/H_{\text{can}}$$

the projection onto the quotient by H_{can}. The quotient X/H_{can} over M is a deformation of $X_o^{(p)}/k$, so its "classifying map" is a morphism of formal Lie varieties

$$\Phi_{\text{can}} : M \to M^{(\sigma)},$$

"defined by" an isomorphism of formal abelian schemes over M

$$\Phi^*_{\text{can}}(X^{(\sigma)}) \cong X/H_{\text{can}}.$$

Thus Φ_{can} is a lifting of the absolute Frobenius endomorphism of $M \otimes k$, and

$$\begin{array}{ccc} F_{\text{can}} : X & \to & X/H_{\text{can}} \cong \Phi^*_{\text{can}}(X^{(\sigma)}) \\ & \searrow & \swarrow \\ & M & \end{array}$$

is a lifting of the relative (to $M \otimes k$) Frobenius endomorphism of $X \otimes k/M \otimes k$. Therefore this morphism induces on H^1_{DR} an M-linear map

$$F^*_{\text{can}} : \Phi^*_{\text{can}} \sigma^* H^1_{\text{DR}}(X/M) \to H^1_{\text{DR}}(X/M)$$

which is none other than the crystalline map $F(\Phi_{\text{can}})$. Because F_{can} is

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a physical morphism, the induced map F^*_{can} respects the Hodge filtration of H^1_{DR}'s. Therefore we have

$$F(\Phi_{\mathrm{can}})(\Phi^*_{\mathrm{can}}(\mathrm{Fil}^1)^{(\sigma)}) \subset \mathrm{Fil}^1 .$$

THEOREM A3.1. The structure of group imposed on M by the Serre-Tate description of M as

$$\mathrm{Ext}(X_o(p^\infty)^{\mathrm{et}}, X_o(p^\infty)^{\mathrm{conn}})$$

coincides with the structure of group on M for which the $q_{\mathrm{ij}} = \exp(\tau_{\mathrm{ij}})$ define an isomorphism of groups

$$M \cong \prod_{i,j} \widehat{G}_m.$$

PROOF. The morphism $\Phi_{\mathrm{can}} : M \to M^{(\sigma)}$ is also a group homomorphism for the Serre-Tate group structure on M. Q.E.D.

A4. FORMAL MODULI OF ORDINARY K3 SURFACES ; SHARPENING OF 2.1.7, 2.1.14

COROLLARY A4.1 (of 2.1.7, 2.1.14). With the hypotheses and notations of 2.1.7 and 2.1.14, there exists a unique morphism

$$\Phi_{\mathrm{can}} : \mathrm{Spf}(A) \to \mathrm{Spf}(A)^{(\sigma)}$$

lifting Frobenius for which the induced crystalline map $F(\Phi_{\mathrm{can}})$ on $H^2_{\mathrm{DR}}(X/A)$

$$F(\Phi_{\mathrm{can}}) : \Phi^*_{\mathrm{can}}\sigma^* H^2_{\mathrm{DR}}(X/A) \to H^2_{\mathrm{DR}}(X/A)$$

preserves the Hodge filtration, i.e. satisfies

$$\begin{cases} F(\Phi_{\mathrm{can}})\Phi^*_{\mathrm{can}}((\mathrm{Fil}^2)^{(\sigma)}) \subset \mathrm{Fil}^2 \\ F(\Phi_{\mathrm{can}})\Phi^*_{\mathrm{can}}((\mathrm{Fil}^2)^{(\sigma)}) \subset \mathrm{Fil}^1 . \end{cases}$$

The group structure on $\mathrm{Spf}(A)$ defined by $q_1, \ldots, q_{20}$ is the unique one for which Φ_{can} is a group homomorphism.

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PROOF. The proof of 2.1.7 shows that any Φ_{can} whose associated $F(\Phi_{\text{can}})$ preserves the Hodge filtration satisfies

$$\Phi_{\text{can}}^{*}(q_i^{(\sigma)}) = (q_i)^p \quad \text{for} \quad i = 1, \dots, 20.$$

Therefore Φ_{can} is a group homomorphism; as it is completely specified by its effect on the q_i, it is unique. Part (iii) of 2.1.7 shows that such a Φ_{can}, preserving the Hodge filtration, does in fact exist. Q.E.D.